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On computing quantum waves exactly from classical action

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We show that the Schrödinger equation can be solved exactly based only on classical least action. Fundamental postulates of quantum mechanics can in turn be derived directly from this construction. The results extend to the relativistic Klein-Gordon, Pauli, and Dirac equations, and suggest a smooth transition between physics across scales. Most quantum mechanics problems have classical versions which involve multiple least action solutions. The associated classical multipaths stem either from the initial position or momentum distribution, or from branch points, generated, e.g. by a multiply connected manifold (double slit experiment), by spatial inequality constraints (particle in a box), or by a singularity (Coulomb potential). We show that the exact Schrödinger wave function ψ can be constructed by combining this classical multi-valued action ϕ with the classical density ρ , computed analytically from ϕ along each extremal action path. The construction is general and does not involve any semi-classical approximation. Quantum wave collapse at measurement can be derived from the classical density change. Entanglement corresponds to a sum of classical particle actions mapping to a tensor product of spinors. The results also provide a simpler computational alternative to Feynman path integrals, as they use only a minimal subset of classical paths.

1. Introduction

Attempts to bridge the conceptual gap between classical and quantum physics have a long and very distinguished history. Central among those is the path-integral formulation of quantum mechanics, starting with Wiener's

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work on stochastic processes, Dirac's discussion of the relation of classical least action to quantum mechanics [1], Feynman's fundamental paper [2,3] on path integral computation, and more recent important extensions such as Duru and Kleinert's time reparametrization [4,5]. Related work includes hydrodynamic [6] and pilot wave [7] interpretations, quasi- and semi-classical approximations [8–11], as well as expansions of quantum commutators based on classical Poisson brackets [12,13]. This paper stems from the same general motivation, but aims to create an *exact* and practical construction of Schrödinger's wave function based solely on classical action.

Section 2 analyzes when classical action solutions of the Lagrangian optimization [14] are multi-valued, due to multiple initial conditions or due to branch points of the associated Hamilton-Jacobi p.d.e. Branch points can arise, e.g. from multiply connected manifolds, spatial inequality constraints, or singularities of the Hamiltonian. In the double slit experiment, for instance, the two-connected manifold implies a two-valued action solution corresponding to the two shortest connections through the slits. Similarly, for a particle in a box, multiple reflections on the walls with different initial velocities induce multiple local minima of the action.

Recall that the classical motion of a physical system corresponds to a local extremum over variational paths $\mathbf{x}(t) = (x^1, \dots, x^N) \in \mathbb{R}^N$ of the system's action,

$$\phi(\mathbf{x}, t, \mathbf{x}_0 \vee \mathbf{p}_0) = \text{extremum}_{\mathbf{x}(t)} \int L(\dot{\mathbf{x}}(t), \mathbf{x}(t), t) dt \quad \text{and} \quad L = \frac{1}{2} \dot{\mathbf{x}}^T \mathbf{M} \dot{\mathbf{x}} + \mathbf{A}^T \mathbf{Q} \dot{\mathbf{x}} - V \quad (1.1)$$

with final position $\mathbf{x}$ at time t , initial position $\mathbf{x}_0$ or (exclusive $\vee$) initial momentum $\mathbf{p}_0$, Lagrangian L , inertia tensor or metric $\mathbf{M}(\mathbf{x})$, potential energy $V(\mathbf{x}, t)$, vector potential $\mathbf{A}(\mathbf{x}, t)$, diagonal charge $\mathbf{Q}$ matrix with the constant charge Q of each particle on the diagonal, e.g. [3,14,15]. The extremal action field $\phi(\mathbf{x}, t)$ can be computed from the Hamilton-Jacobi p.d.e. [16–18], with H piecewise $C^2(\mathbf{x}, t, \nabla\phi)$,

$$-\frac{\partial\phi}{\partial t} = H = \frac{1}{2}(\nabla\phi - \mathbf{Q}\mathbf{A})^T \mathbf{M}^{-1}(\nabla\phi - \mathbf{Q}\mathbf{A}) + V. \quad (1.2)$$

In this paper, unless otherwise specified, we will simply use the term action to refer to such local extremal action (or local stationary action).

The symmetric metric $\mathbf{M}(\mathbf{x})$ is required to be uniformly invertible, but is not necessarily positive definite. We use the standard Laplace-Beltrami tensor operator [15,19,20]

$$\left. \begin{aligned} \nabla_{\mathbf{M}} \cdot \mathbf{f} &= \frac{1}{\sqrt{\det \mathbf{M}}} \sum_{n=1}^N \frac{\partial}{\partial x^n} (\sqrt{\det \mathbf{M}} f^n) \quad \text{for } \mathbf{f}(\mathbf{x}, t) = (f^1, \dots, f^N) \in \mathbb{R}^N \\ \Delta_{\mathbf{M}} f &= \nabla_{\mathbf{M}} \cdot (\mathbf{M}^{-1} \nabla f), \quad \nabla f = \frac{\partial f}{\partial \mathbf{x}} \quad \text{for } f(\mathbf{x}, t) \in \mathbb{R} \end{aligned} \right\} \quad (1.3)$$

for a given metric $\mathbf{M}(\mathbf{x})$. No index is used for $\mathbf{M}(\mathbf{x}) = \mathbf{I}$. The vector potential $\mathbf{A}(\mathbf{x}, t)$ is assumed to follow the Coulomb or Lorenz gauge $\nabla_{\mathbf{M}} \cdot \mathbf{A} = 0$ [21,22].

Section 3 shows that this multi-valued action can be converted exactly into the quantum wave function, provided it is weighted along each branch according to the classical *density* of the velocity field implied by the action. A key point is that the classical density ρ can be computed analytically along each action branch simply by using the classical continuity equation [23],

$$0 = \frac{\partial}{\partial t} \rho + \nabla_{\mathbf{M}} \cdot (\rho \dot{\mathbf{x}}) = \frac{d\rho}{dt} + \rho \nabla_{\mathbf{M}} \cdot \dot{\mathbf{x}}. \quad (1.4)$$

We will see that this evolution of the classical density distribution over time implies the time evolution of the quantum probability density distribution over time.

The above construction is general and does not involve any quasi- or semi-classical approximation. Dirac [1] introduced an approximate relation between the single-valued action (1.1) and the quantum wave ψ of the Schrödinger equation,

$$\psi \approx e^{\frac{i}{\hbar} \phi(\mathbf{x}, t)} \quad (1.5)$$

under the assumption $\hbar\Delta_{\mathbf{M}}\phi \approx 0$, with $\hbar$ the reduced Planck constant [24]. Under the same assumption, more general quasi-classical expansions [8] later used the 1-dimensional approximation $\psi \approx C_1/\sqrt{|\nabla\phi|} e^{i/\hbar\phi} + C_2/\sqrt{|\nabla\phi|} e^{-i/\hbar\phi}$ where C_1, C_2 are constants. Another semi-classical approximation, $\psi(\mathbf{x}, \mathbf{x}_0, t) \approx \sum_{\text{class. paths } j} (2\pi i\hbar)^{-N/2} |\det(\partial^2\phi_j/\partial\mathbf{x}\partial\mathbf{x}_0)|^{1/2} e^{i/\hbar\phi_j - \frac{i\pi}{2}v_j}$, was derived from the Van Vleck determinant [9] and the associated Maslov indices [10], which count the number of times v_j each classical path crosses a caustic and introduces accordingly a phase correction $-\frac{\pi}{2}v_j$. All semi-classical approximations become quite incorrect for small masses M , close to constraints or to singular potential fields, where actually $\Delta_{\mathbf{M}}\phi$ can become unbounded, reflecting the rapid change in local momentum.

As detailed in §3, semi-classical approximations can actually be replaced by an exact result, by substituting the full set of extremal actions ϕ_j for Dirac's action ϕ in (1.5), and *weighting* each resulting term $e^{i/\hbar\phi_j}$ by the square root of the classical density ρ_j , computed from (1.4) along that action branch. In other words, Feynman's infinity of non-classical zig-zag paths can be reduced to the subset of all extremal classical action paths, each weighted according to the classical fluid density computed along the path. This yields an exact construction of the wave function $\psi(\mathbf{x}, t)$ of the Schrödinger equation [25,26],

$$0 = \left[\frac{\hbar}{i} \frac{\partial}{\partial t} + \frac{1}{2} \left(\frac{\hbar}{i} \nabla_{\mathbf{M}} - \mathbf{Q}\mathbf{A} \right) \cdot \mathbf{M}^{-1} \left(\frac{\hbar}{i} \nabla - \mathbf{Q}\mathbf{A} \right) + V \right] \psi. \quad (1.6)$$

Examples illustrate the computation of the wave function based on the set of all classical action solutions and their associated classical densities. They include the double slit experiment, the Aharonov–Bohm effect, a particle in a box, quantum tunnelling, a harmonic oscillator, and the Coulomb potential of a hydrogen atom and its relation to Kepler orbits. We also show how wave collapse at measurement [27] is a consequence of our formalism.

Section 4 extends the results to the relativistic Klein-Gordon [28–30], Dirac, Pauli [31,32] and Maxwell equations [33]. Illustrative examples include the Einstein-Podolsky-Rosen experiment with entangled spinning particles, and positron-electron creation in quantum electrodynamics.

Concluding remarks and perspectives are offered in §5.

2. Multi-valued local least action and multipaths

Rather than being based on an approximation or a mathematical expansion, our mapping from classical to quantum mechanics relies on a different and exact mechanism. This mechanism is built on the combination of an exhaustive census of the multiple classical action solutions, which we first discuss, and of a key algebraic result relating the solutions of the Schrödinger equation to those of the Hamilton-Jacobi p.d.e. (§3, lemma 3.1).

Let us first introduce spatial inequality constraints on a multiply connected manifold, and then define action branches on this manifold.

Definition 2.1. The constrained multiply connected manifold $\mathbb{G}^N \subseteq \mathbb{R}^N$ is defined by the $g = 1, \dots, G$ inequality constraints $f_g(\mathbf{x}, t) \leq 0$. The set of active constraints $\mathbb{G}(\mathbf{x}, t) \subseteq \{1, \dots, G\}$ is the set of indices g on the boundary $\partial\mathbb{G}^N$ of $\mathbb{G}^N$ such that $f_g(\mathbf{x}, t) = 0$.

At the time of reflection on $\partial\mathbb{G}^N$, the kinetic energy is preserved for a fully elastic reflection, whereas it vanishes for a plastic collision.

Definition 2.2. The action branch set is the set of local extremal action fields $\phi_j(\mathbf{x}, t)$, $\mathbf{x} \in \mathbb{G}^N$, $t \geq 0$ which are different at least at one $(\mathbf{x}, t)$, except for the integration constant for each given initial condition in (1.1).

Note that by this definition, a single action branch $\phi_j(\mathbf{x}, t)$ always spans the complete constrained manifold $\mathbb{G}^N$ even if it is partially identical to another action branch. Also, omitting a branch j in the calculation of the multi-valued action is equivalent to assuming that the initial density ρ_j of this branch is zero.

Multi-valued least action branches naturally arise from multiple initial conditions in $\mathbf{x}_0$ or $\mathbf{p}_0$ in (1.1). After the initialization, they can also arise at branch points.

Definition 2.3. The set of branch points $\mathbb{B}^N \subseteq \mathbb{G}^N$ consists of all points where the set of distinct $\phi_j(\mathbf{x}, t)$ changes, for $t \geq 0$. Each branch point $\mathbf{x}$ occurs $(b_{\mathbf{x}} - 1)$ times in $\mathbb{B}^N$, where $b_{\mathbf{x}} \in \mathbb{N}$ is the number of distinct action solutions in the local neighbourhood of $\mathbf{x}$.

The extremization (1.1) and the associated Euler-Lagrange and Hamilton o.d.e. for $\mathbf{x} \in \mathbb{R}^N$ can be extended to the case of constrained positions $\mathbf{x} \in \mathbb{G}^N \subset \mathbb{R}^N$ of definition 2.1. The action ϕ of (1.1) has a local extremum [14] if the variation of the action (1.1)

$$\begin{aligned} \delta\phi &= \int_0^t \frac{\partial L}{\partial \dot{\mathbf{x}}} \delta \dot{\mathbf{x}} + \frac{\partial L}{\partial \mathbf{x}} \delta \mathbf{x} d\theta \\ &= \left[\frac{\partial L}{\partial \dot{\mathbf{x}}} \delta \mathbf{x} \right]_0^t - \int_0^t \left(\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{x}}} - \frac{\partial L}{\partial \mathbf{x}} \right) \delta \mathbf{x} d\theta = \int_0^t \sum_{g \in \mathbb{G}} \lambda_g \frac{\partial f_g}{\partial \mathbf{x}} \delta \mathbf{x} d\theta \end{aligned}$$

is only non-zero orthogonal to an active constraint, where the Lagrange parameter λ_g defines the magnitude of the cost gradient at the active constraint. The first term on the right-hand side is zero since $\delta \mathbf{x}$ is zero at the start and end points. In between $\delta \mathbf{x}$ can take on any arbitrary value. Thus a local least action solution satisfies

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\mathbf{x}}} - \frac{\partial L}{\partial \mathbf{x}} = \frac{d\nabla\phi}{dt} + \frac{\partial H}{\partial \mathbf{x}} = \sum_{g \in \mathbb{G}} \lambda_g \frac{\partial f_g}{\partial \mathbf{x}}. \quad (2.1)$$

This extends the usual Euler-Lagrange or Hamilton's o.d.e. (e.g. [14,16]) with Lagrangian collision forces activated by inequality constraints.

The existence and uniqueness of a solution of (2.1) is guaranteed by the Picard-Lindelöf Theorem [34] for Lipschitz [35] continuous $-\partial H/\partial \mathbf{x} + \sum_{g \in \mathbb{G}} \lambda_g \frac{\partial f_g}{\partial \mathbf{x}}$. Contraction theory [36] guarantees a bounded contraction rate of the Hamilton-Jacobi p.d.e. (1.2) for bounded $\Delta_{\mathbf{M}}\phi_j(\mathbf{x} \in \mathbb{G}^N, t)$ and $\nabla\phi_j(\mathbf{x} \in \partial\mathbb{G}^N, t)$ [37]. For a given initial condition, a bounded contraction rate in turn implies the existence and uniqueness of a solution of the Hamilton-Jacobi p.d.e. (1.2). Thus, the branch points of definition 2.3 can only occur at unbounded $\Delta_{\mathbf{M}}\phi_j(\mathbf{x} \in \mathbb{G}^N, t)$ or unbounded $\nabla\phi_j(\mathbf{x} \in \partial\mathbb{G}^N, t)$.

These properties and (1.4) may be summarized as follows, introducing for generality an ensemble $\mathbb{E}$ of possible initial density conditions to describe different initial conditions occurring with given probabilities.

Theorem 2.4. The multi-valued least action field $\phi_j(\mathbf{x}, t, \mathbf{x}_0 \vee \mathbf{p}_0)$, with H piecewise $\mathbf{C}^2(\mathbf{x}, t, \nabla\phi_j)$,

$$-\frac{\partial\phi_j}{\partial t} = H = \frac{1}{2}(\nabla\phi_j - \mathbf{QA})^T \mathbf{M}^{-1}(\nabla\phi_j - \mathbf{QA}) + V \quad \forall t \geq 0 \quad (2.2)$$

locally extremizes (1.1) on the constrained multiply connected manifold $\mathbf{x} \in \mathbb{G}^N$ of definition 2.1, yielding the multipaths

$$\frac{d\nabla\phi_j}{dt} + \frac{\partial H}{\partial \mathbf{x}} = \sum_{g \in \mathbb{G}} \frac{\partial f_g}{\partial \mathbf{x}} \lambda_g \quad (2.3)$$

and

$$\mathbf{M}(\mathbf{x}) \frac{d\mathbf{x}}{dt} = \nabla\phi_j - \mathbf{QA} \quad (2.4)$$

where the reflection force λ_g fulfills definition 2.1.

The action is said to be $\mathbb{J}$ -valued, where $\mathbb{J}$ indexes the set

$$\{\mathbf{x}_0 \vee \mathbf{p}_0\} \times \mathbb{B}^N \quad (2.5)$$

which accounts for all initial conditions in $\mathbf{x}_0$ or $\mathbf{p}_0$ in (1.1) and all branch points $\mathbb{B}^N$ of definition 2.3. Branch points exist only at unbounded $\Delta_{\mathbf{M}}\phi_j(\mathbf{x} \in \mathbb{G}^N, t)$ or unbounded $\nabla\phi_j(\mathbf{x} \in \partial\mathbb{G}^N, t)$. $\mathbb{B}$ indexes all

branch points $\mathbb{B}^N$. Each element of $\mathbb{J}$ corresponds to an action branch ϕ_j , while each element of $\mathbb{B}$ corresponds to an action branch ϕ_j for a specific initial condition $\{\mathbf{x}_0 \searrow \mathbf{p}_0\}$.

Combining (1.4), (2.4) the classical density can be computed along each extremal path $\mathbf{x}_j(t)$, yielding the path integral

$$\rho_j^\epsilon(\mathbf{x}_j(t), t) = \rho_{oj}^\epsilon e^{-\int_0^t \Delta_M \phi_j(\mathbf{x}(\theta), \theta) d\theta} \quad (2.6)$$

where the gauge $\nabla_M \cdot \mathbf{A} = 0$ was used. The initial density $\rho_{oj}^\epsilon = \rho_j^\epsilon(\mathbf{x}_0 \searrow \mathbf{p}_0, 0)$ is normalized to the real probability $\int_{\mathbb{G}^N} \rho_{oj}^\epsilon dx^1 \dots dx^N = p^\epsilon \geq 0$ of each element ϵ in an ensemble $\mathbb{E}$, with $\sum_{\epsilon \in \mathbb{E}} p^\epsilon = 1$. The density $\rho_j^\epsilon(\mathbf{x}_j(t), t)$ corresponds to the classical probability density to find the Hamiltonian system at $\mathbf{x}_j$ at time t . The superscript ϵ is removed if the ensemble has only one element.

Theorem 2.4 computes the $\mathbb{B}$ multipaths connecting an initial $\mathbf{x}_0 \searrow \mathbf{p}_0$ to a final $\mathbf{x}$. Its result is *determined* in that sense, but in contrast to Newton's law [38] it is in general *not deterministic* since the initial condition is incomplete, as initially only $\mathbf{x}_0$ or $\mathbf{p}_0$ can be defined. The continuity equation (2.6) is a *classical stochastic description* of how an initial probability density distribution propagates in the future, although no noise source exists in the Hamiltonian (2.2) beyond the non-Lipschitz constraint force in (2.3). Hence the incompleteness of the initial condition leads to a classical stochastic density field $\rho_j(\mathbf{x}_j, t)$ from an initial $\mathbf{x}_0 \searrow \mathbf{p}_0$ to any final $\mathbf{x} \in \mathbb{G}^N$. Finally, the original derivation by Hamilton & Jacobi [17,18], defined on a manifold $\mathbb{R}^N$ rather than on a constrained manifold $\mathbb{G}^N$, was not formulated to predict multipath solutions beyond the multiplicity of initial conditions in (2.5). We will see that using all these classical features of theorem 2.4, rather than using Newton's single deterministic path [38], allows us to resolve most conflicts of quantum physics with classical physics.

3. Exact wave computation from classical multi-valued action and density

We now build on the above result to show that the Schrödinger equation can be solved exactly by computing wave functions directly from the Hamilton-Jacobi p.d.e. At the time of Schrödinger and Feynman, the Hamilton-Jacobi p.d.e. only had a single-valued least action ϕ for a given initial condition. Hence the Feynman path integral [3] had to consider all non-classical stochastic zig-zag paths with a time slicing approach, rather than just those minimizing (1.1). This stochastic process noise along the path can be avoided if one uses the determined multi-valued local least action solutions of theorem 2.4. As we now show, the Schrödinger equation (1.6) can be solved on each extremal branch j of definition 2.2 by

$$\psi_j = \sqrt{\rho_j} e^{\frac{i}{\hbar} \phi_j} = \sqrt{\rho_{oj}}(\mathbf{x}_0 \searrow \mathbf{p}_0, 0) e^{-1/2 \int_0^t \Delta_M \phi_j(\mathbf{x}(\theta), \theta) d\theta + \frac{i}{\hbar} \phi_j} \quad (3.1)$$

using the action field $\phi_j(\mathbf{x}, t, \mathbf{x}_0 \searrow \mathbf{p}_0)$ and the classical density path integral (2.6) of theorem 2.4. Taking the sum of the waves ψ_j over all branches j then yields the overall wave ψ .

Lemma 3.1. For each branch j , plugging the piecewise $\mathcal{C}^2$ wave ψ_j from (3.1) into the Schrödinger equation (1.6) exactly leads to the Hamilton-Jacobi p.d.e. (2.2),

$$\begin{aligned} & \left[\frac{\hbar}{i} \frac{\partial}{\partial t} + \frac{1}{2} \left(\frac{\hbar}{i} \nabla_M - \mathbf{QA} \right) \cdot \mathbf{M}^{-1} \left(\frac{\hbar}{i} \nabla - \mathbf{QA} \right) + V \right] \psi_j \\ &= \left[\frac{\partial \phi_j}{\partial t} + \frac{1}{2} (\nabla \phi_j - \mathbf{QA})^T \mathbf{M}^{-1} (\nabla \phi_j - \mathbf{QA}) + V \right] \psi_j = 0 \end{aligned} \quad (3.2)$$

The first equation is an operator equation, which becomes a product in the second equation thanks to the exponential form of the wave ψ_j . Equation (3.2) holds for piecewise $\mathcal{C}^2$ real, complex or quaternion actions and waves.

Proof. The density (2.6) is only defined along each individual path $\mathbf{x}(t)$ (2.4), so that its total differential has no variation with respect to $\mathbf{x}$. Replacing ψ_j in the first line of (3.2) yields

$$\left[\frac{\partial \phi_j}{\partial t} - \frac{\hbar}{2i} \Delta_{\mathbf{M}} \phi_j + \frac{1}{2} (\nabla \phi_j - \mathbf{Q}\mathbf{A})^T \mathbf{M}^{-1} (\nabla \phi_j - \mathbf{Q}\mathbf{A}) + \frac{\hbar}{2i} \Delta_{\mathbf{M}} \phi_j \right] \psi_j$$

which in turn yields the second line. ■

In earlier attempts [6,9,10,39] to map the Hamilton-Jacobi equation to the Schrödinger equation, (3.2) needs to be extended by an extra error term, since the normalization factor is based on a *general field* $R(\mathbf{x}, t)$, not necessarily fulfilling the continuity equation (1.4). Instead, for a given $\mathbf{x}_0$ or $\mathbf{p}_0$ at $t = 0$, the proposed density path integral $\rho_j(\mathbf{x}(t), t)$ of (2.6) fulfills (1.4) and depends only on t along the path $\mathbf{x}(t)$. Note that both classical density and classical action are path integrals, where the density is path-dependent and the action is path-independent. Also, earlier attempts do not always use a complete $\mathbb{J}$ -valued action of (2.5). In this paper we may use interchangeably the notations $\sum$ and $\int$ in normalized sums as some indices may include mixtures of discrete and continuous quantities.

Theorem 3.2. *The wave function $\psi^\epsilon(\mathbf{x}, t)$ of the Schrödinger equation (1.6) can be computed from the multi-valued least action field $\phi_j(\mathbf{x}, t, \mathbf{x}_0 \vee \mathbf{p}_0)$, the initial action field $\phi_{oj}(\mathbf{x}, 0, \mathbf{x}_0 \vee \mathbf{p}_0)$, the classical density path integral (2.6) of the ensemble $\mathbb{E}$ of theorem 2.4*

$$\psi^\epsilon = \sum_{j \in \mathbb{J}} \sqrt{\rho_j^\epsilon} e^{\frac{i}{\hbar} \phi_j}. \quad (3.3)$$

For each j , the action provides the phase of the wave and the square root density normalized as above gives the corresponding gain. Equivalently, $\psi^\epsilon(\mathbf{x}, t)$ can be expressed using the Feynman kernel $K_j(\mathbf{x}, t, \mathbf{x}_0 \vee \mathbf{p}_0)$ [3] computed from classical action and density,

$$\psi^\epsilon = \sum_{j \in \mathbb{J}} K_j \psi_{oj}^\epsilon \quad \text{where} \quad K_j = \sqrt{\frac{\rho_j}{\rho_{oj}}} e^{\frac{i}{\hbar} (\phi_j - \phi_{oj})} \quad \text{and} \quad \psi_{oj}^\epsilon = \sqrt{\rho_{oj}^\epsilon} e^{\frac{i}{\hbar} \phi_{oj}} \quad (3.4)$$

The associated quantum density matrix at time t

$$\varrho(\mathbf{x}, t) = \sum_{\epsilon \in \mathbb{E}} p^\epsilon \psi^\epsilon \psi^{\epsilon\dagger} \quad (3.5)$$

is the determined forward mapping along all classical paths of theorem 2.4 from the initial quantum density distribution at $t = 0$.

Note that an initially normalized distribution (2.6, 3.3) remains normalized $\forall t \geq 0$, as (1.6) and its conjugate imply that

$$\frac{\partial}{\partial t} \int_{\mathbb{G}^n} \psi^{\epsilon\dagger} \psi^\epsilon \, d\mathbf{x}^1 \dots d\mathbf{x}^N = 0. \quad (3.6)$$

Also note that theorems 2.4 and 3.2 immediately extend to complex or quaternion actions, as long as a real classical path can be constructed, for instance using superposition. Equation (3.3) of theorem 3.2

- replaces Dirac's wave approximation (1.5) and approximate quasi-classical expansions [8] by an exact computation, for any action with $\hbar \Delta_{\mathbf{M}} \phi \not\approx 0$. Note that $\Delta_{\mathbf{M}} \phi$ of (1.3) can become large or even unbounded for small M close to constraints or singularities, i.e. in regions where most quantum phenomena occur.

- uses only the $\mathbb{J}$ -valued classical multipaths or actions from theorem 2.4, which are a subset of all zig-zag paths in Feynman's path integral

$$\psi(\mathbf{x}_o, \mathbf{x}, t) = \frac{1}{Z} \int_{\mathbf{x}_o}^{\mathbf{x}} e^{\frac{i}{\hbar} \int_0^t L d\theta} \mathcal{D}\mathbf{x} \quad (3.7)$$

where $\mathcal{D}\mathbf{x}$ denotes the integration over ∞^∞ stochastically time-sliced zig-zag paths and Z is the normalization factor [3]. This also applies to more recent important developments such as Duru and Kleinert's time reparameterization [4,5,40–42].

- extends Feynman's key result on Gaussian integrals of quadratic least actions in $\mathbb{R}^N$ [3] to general least actions in a constrained subset $\mathbb{G}^N \subseteq \mathbb{R}^N$.

Wave collapse at measurement [25,27] is a natural consequence of our formulation. It can be derived directly from the change of the classical density $\sqrt{\rho_j}(\mathbf{x})$ in theorem 2.4 due to a classical measurement, as we now discuss. Intuitively, a measurement transforms the classical density distribution simply into a Dirac distribution in measurement coordinates, leading to the determination of $\mathbb{J}$ classical multipaths (2.5) from $\{\mathbf{x}_o \vee \mathbf{p}_o\}$ to that measurement, and thus through (3.3) to a wave collapse in the corresponding ψ .

Let us map the classical density $\sqrt{\rho_j}(\mathbf{x})$ to $\sqrt{\rho_j}(\mathbf{y}) = U(\mathbf{x}, \mathbf{y}) = \frac{1}{\sqrt{2\pi\hbar}} e^{-\frac{i\mathbf{y}\mathbf{x}}{\hbar}}$ [22], where the operator $\hat{U}$ is unitary. We use the notation $\hat{U}a = \int_{\mathbf{x} \in \mathbb{G}^N} U(\mathbf{x}, \mathbf{y}) a(\mathbf{x}) d\mathbf{x}^1, \dots, d\mathbf{x}^N$ with $\mathbf{y} \in \mathbb{R}^N$, where for instance $U(\mathbf{x}, \mathbf{y}) = \delta(\mathbf{x} - \mathbf{y})$ implies a spatial density distribution $\sqrt{\rho_j}(\mathbf{y}) = \sqrt{\rho_j}(\mathbf{x})$, whereas $U(\mathbf{x}, \mathbf{y}) = 1/\sqrt{2\pi\hbar} e^{-\frac{i\mathbf{y}\mathbf{x}}{\hbar}}$ yields the Fourier transform and thus implies a momentum $\mathbf{y}$ density distribution.

Consider now a classical measurement $\mathbf{y}_k$ of $\mathbf{y}$ in the density distribution $\sqrt{\rho_j}(\mathbf{y})$. Since this classical measurement tells us the exact $\mathbf{y}$, which was not known before the measurement, it turns the general classical density distribution $\sqrt{\rho_j}(\mathbf{y})$ before the measurement into the Dirac impulse $\sqrt{P_k}(\mathbf{y}) = \hat{U}\sqrt{P_k}(\mathbf{x}) = \delta(\mathbf{y} - \mathbf{y}_k)$ after the measurement. From (3.3), this change of the classical density in turn implies that at the measurement, a general wave collapses into a wave $\psi_k(\mathbf{y}) = \hat{U}\psi_k(\mathbf{x}) = \delta(\mathbf{y} - \mathbf{y}_k) \sum_{j \in \mathbb{J}} e^{i/h\phi_j}$. The factor $\sum_{j \in \mathbb{J}} e^{i/h\phi_j}$ disappears when $\psi_k(\mathbf{y})$ is normalized.

As discussed in §2, the Hamilton-Jacobi formulation in theorem 2.4 determines the $\mathbb{J}$ multipaths (2.5) to all final positions $\mathbf{x} \in \mathbb{G}^N$ from either an initial $\mathbf{x}_o$ or an initial $\mathbf{p}_o$. This formulation is not deterministic, since at the initialization either $\mathbf{p}_o$ or $\mathbf{x}_o$ is undefined. This classical initial ambiguity is resolved when the final $\mathbf{y}_k$ is measured, i.e. at the measurement time the $\mathbb{J}$ classical multipaths (2.5) from $\{\mathbf{x}_o \vee \mathbf{p}_o\}$ to $\mathbf{y}_k$ are determined.

Lemma 3.3. Consider a Hermitian quantum measurement operator $\hat{Y} = \hat{U}^\dagger \text{diag}(\mathbf{y}) \hat{U}$, based on a measurement $\mathbf{y} \in \mathbb{R}^N$ and a unitary operator $\hat{U}(\mathbf{x}, \mathbf{y})$. Let $\sqrt{\rho_j}(\mathbf{x})$ be the classical density distribution at measurement time t , with $\mathbf{x} \in \mathbb{G}^N$. Mapping $\sqrt{\rho_j}(\mathbf{x})$ to $\sqrt{\rho_j}(\mathbf{y}) = \hat{U}\sqrt{\rho_j}(\mathbf{x})$, a specific measurement result $\mathbf{y}_k$ of $\mathbf{y}$ implies that

- the classical square root density distribution $\sqrt{\rho_j}(\mathbf{y})$ before the measurement $\mathbf{y}$ turns into $\sqrt{P_k}(\mathbf{y}) = \hat{U}\sqrt{P_k}(\mathbf{x}) = \delta(\mathbf{y} - \mathbf{y}_k)$ after the measurement. Since $\sqrt{\rho_j}(\mathbf{y})$ is constrained to $\mathbf{y}_k$ at the measurement, $\sqrt{P_k}(\mathbf{y}) = \delta(\mathbf{y} - \mathbf{y}_k)$ is a branch point of theorem 2.4.
- the quantum wave $\psi(\mathbf{y})$ before the measurement collapses into the normalized eigenwave $\psi_k(\mathbf{y}) = \hat{U}\psi_k(\mathbf{x}) = \delta(\mathbf{y} - \mathbf{y}_k)$. The wave $\psi_k(\mathbf{y})$ is a solution of $\text{diag}(\mathbf{y})\psi_k(\mathbf{y}) = \text{diag}(\mathbf{y}_k)\psi_k(\mathbf{y})$, which can be written equivalently as $\hat{Y}\psi_k(\mathbf{x}) = \text{diag}(\mathbf{y}_k)\psi_k(\mathbf{x})$.

At measurement time, the initial condition in $\mathbf{x}_o$ and $\mathbf{p}_o$ of theorem 2.4 is completed by the measurement $\mathbf{y}_k$, which determines in theorem 2.4 the $\mathbb{J}$ classical multipaths (2.5) from $\{\mathbf{x}_o \vee \mathbf{p}_o\}$ to $\mathbf{y}_k$.

In the formulation above, all eigenvalues are simple. Multiple eigenvalues can occur if $\dim \mathbf{y} < N$. For instance, the temporal operator $U(t, y) = 1/\sqrt{2\pi\hbar} e^{-ity/\hbar}$ generates the temporal Fourier transform, and thus yields the density distribution of a scalar energy y [3]. The classical density

becomes $\sqrt{P_k}(y) = \hat{U}\sqrt{P_k}(x) = \delta(y - y_k)$ after the measurement, and the wave collapses to the eigenwave $\Psi_k(y) = \hat{U}\Psi_k(x) = \delta(y - y_k)$. In this case, multiple eigenwaves $\Psi_k(x)$ correspond to the same eigenvalue y .

Note that theorem 3.2 and lemma 3.3 directly derive fundamental postulates of quantum mechanics [25] from the classical physics in theorem 2.4. Namely, (i) the quantum system state is represented as a wave function $\psi_j(x, t)$, now derived from the classical action $\phi_j(x, t)$, (ii) the wave function solves the Schrödinger equation (both from lemma 3.1 and theorem 3.2), (iii) which implies wave collapse at measurements (from lemma 3.3).

Born's measurement rule [25,27] remains a postulate. It states that the probability to measure the position x is the (now classically derived) scalar quantum density $\varrho(x, t)$ of (3.5) in theorem 3.2, or in the matrix case the diagonal elements of $\varrho(x, t)$. It implies that the probability of measuring y_k in lemma 3.3 from the wave $\psi(y) = \sum_{k \in \mathbb{N}} c_k \Psi_k(y)$ is $\varrho_{y_k} = \sum_k$ with same y_k $c_k^* c_k$.

Also note that lemma 3.3 suggests the possibility of a classically-based interpretation of wave collapse at a quantum measurement, as an alternative to the Copenhagen interpretation [26,43–45]. The Copenhagen interpretation suggests that this decision is taken at the measurement time t . The classically-based interpretation would be that these decisions are taken before the measurement, at the initial condition $t = 0$ and at the branch points along the $\mathbb{J}$ multipath. Which interpretation best describes the nature of physical reality remains an open question since both interpretations lead to the same experimental results. However, lemma 3.3 is fully derived from and consistent with the classical Hamilton–Jacobi and Euler formulations of theorem 2.4.

Finally, note that for periodic waves a Bohr-like quantization rule [46] can be derived.

Lemma 3.4. *Given periodic waves $\psi_j = \sqrt{\rho}(x, t) e^{\frac{i}{\hbar}(\phi(x, t) + k\varphi(\omega))}$ in theorem 3.2, with $\mathbb{J} = \{\omega \in \mathbb{R} \times k \in \mathbb{N}\}$ and φ an arbitrary continuous function, the continuous parameter ω in the cumulated wave ψ is quantized as $\frac{\varphi(\omega)}{\hbar} = 2\pi k$.*

Proof. The cumulated wave is derived from the geometric series

$$\lim_{K \rightarrow +\infty} \frac{1}{K} \sum_{k=0}^K e^{ik\frac{\varphi}{\hbar}} = \lim_{K \rightarrow +\infty} \frac{1}{K} \frac{1}{1 - e^{i\frac{\varphi}{\hbar}}} = \begin{cases} 1 & \text{for any } \frac{\varphi}{\hbar} = 2\pi k \\ 0 & \text{for any } \frac{\varphi}{\hbar} \neq 2\pi k \end{cases} \quad k \in \mathbb{N}.$$

All periodic actions are included in the summation, but the above implies that only $2\pi k$ -periodic actions remain. This is the source of quantization. ■

Examples

We illustrate through examples how the wave function can be systematically constructed based only on classical action and classical density. Each example first finds the multi-valued action solution of the classical Hamilton–Jacobi p.d.e. of theorem 2.4. Position and time coordinates are chosen to simplify the derivation. Next, the classical densities are computed for each action branch. Finally, the quantum wave is constructed from theorem 3.2. Basic computational tools are introduced as needed before each example.

Example 3.5 (Double slit experiment). Consider the classical Hamiltonian–Jacobi p.d.e. (2.2) on the two-connected manifold $\mathbb{G}^3$ in figure 1a

$$-\frac{\partial \phi_j}{\partial t} = H = \frac{1}{2M} \nabla \phi_j^T \nabla \phi_j \quad \mathbf{x} = (x^1, x^2, x^3)^T \in \mathbb{G}^3 = \mathbb{R}^3 \setminus \mathbb{W}^3$$

with constant mass M , wall with two holes $\mathbb{W}^3 = \{x^1 = 0\} \setminus \mathbb{H}^3$, where $\mathbb{H}^3 = \{x_1, x_2\}$ with $\mathbf{x}_{1,2} = (0, \pm 5, 0)^T$, and initial momentum p_o of the particle (for instance, an electron). Letting $r_j =$

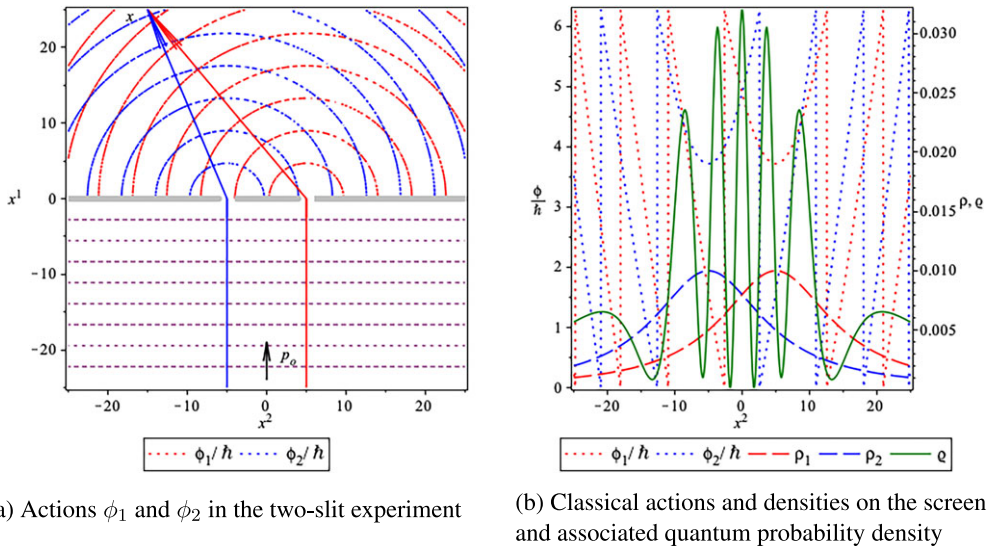


Figure 1. Classical particle paths and quantum wave in the double-slit experiment.

$\sqrt{(x - x_j)^T(x - x_j)}$ after the wall, the $\mathbb{B} = \{1, 2\}$ - valued least action of theorem 2.4 is

$$\phi_j = \begin{cases} p_o x^1 - Et & \text{for } x^1 < 0 \\ p_o r_j - Et & \text{for } x^1 \geq 0 \end{cases} \quad \text{where } E = \frac{p_o^2}{2M}$$

They are shown in figure 1a. Since $H = E$ is constant and there is no potential term, in this example the action simply corresponds to the geometric distance. Using in (1.3) the spherical Laplacian $\Delta_M f(r) = \frac{1}{Mr^2} \frac{\partial}{\partial r} (r^2 \frac{\partial f}{\partial r}) = \frac{1}{Mr} \frac{\partial^2}{\partial r^2} (rf)$, the classical density (2.6) is

$$\Delta_M \phi_j = \begin{cases} 0 & \text{for } x^1 < 0 \\ \frac{2}{r_j} \frac{p_o}{M} = \frac{\dot{r}_j}{r_j} & \text{for } x^1 \geq 0 \end{cases} \implies \rho_j = \begin{cases} 1 & \text{for } x^1 < 0 \\ \frac{1}{r_j^2} & \text{for } x^1 \geq 0. \end{cases} \quad (3.8)$$

The least action branches are illustrated in figure 1a. Both slits are branch points with fully elastic collision forces in (2.3) and an infinite classical density. The classical non-Lipschitz constraint forces (2.3) in the two slits lead to an infinity of radial paths (2.4) from each slit, which connect every measurement pixel x^2 on the screen at $x^1 = 10$ with two least action paths (2.4). Thus, the statistical density distribution $\rho(x(t))$ behind the slits is just the evolution of the Dirac densities in the slits along the density path integrals (2.6) after the slits.

Theorem 3.2 in turn yields the (un-normalized) wave function (3.3),

$$\psi = \sum_{j \in \mathbb{B}} \sqrt{\rho_j} e^{\frac{i}{\hbar} \phi_j} = e^{-\frac{i}{\hbar} Et} \begin{cases} e^{\frac{i}{\hbar} p_o x^1} & \text{for } x^1 < 0 \\ \frac{1}{r_1} e^{\frac{i}{\hbar} p_o r_1} + \frac{1}{r_2} e^{\frac{i}{\hbar} p_o r_2} & \text{for } x^1 \geq 0. \end{cases} \quad (3.9)$$

One can directly confirm that each term fulfills the Schrödinger equation (1.6). When specialized to the far field, this result matches the well-known two-slit Fraunhofer wave function [47]. The wave collapse in both slits can be interpreted according to theorem 3.2 as the transition in figure 1a from the flat least action branch before the wall to the two conic least action branches after the wall.

With $r_j = \sqrt{(x^1)^2 + (x^2 \pm 5)^2}$, the classical actions $\phi_j/\hbar = \frac{p_o r_j}{\hbar} \pmod{2\pi}$ (where the 2π periodic part is removed), densities $\rho_j = r_j^{-2}$ and the resulting near field probability density $\varrho = \psi \psi^\dagger$ on the

screen at $x^1 = 10$ are plotted in [figure 1b](#) with $\frac{p_0}{\hbar} = 2$. The quantum density ϱ is a phase weighted combination of the two classical densities ρ_j , with the two phases derived from the actions. On the left and right sides of the figure the two phases have a nearly constant offset, whereas in the centre the offset changes significantly which leads to the wave oscillation. Feynman's zig-zag path integrals (3.7), which were originally motivated by this example, can now be reduced to just two paths.

The Hamilton-Jacobi formulation of theorem 2.4 cannot deterministically predict from the slits where the particle hits the screen. Indeed, in the slits only the position is given, but not the momentum, which is affected by the non-Lipschitz constraint force. However, the path from the slit to the screen is a determined straight line following (2.4) of theorem 2.4. Lemma 3.3 thus allows a classically-based interpretation that the decision where the particle hits the screen is already taken in the slits from the non-Lipschitz constraint force (2.4) of theorem 2.4, i.e. before the final position is measured on the screen.

Also, measuring the particle position in slit 2 with a photon, for instance, would change the branch index from $j = 1, 2$ to $j = 2$, leading to a single classical action cone behind the wall. Note that in principle this example extends to slits or holes of finite width in $\mathbb{H}^3$, extending the summation in (3.9) to all elements of $\mathbb{H}^3$. A particle described, e.g. with Laguerre polynomials [48] will always collide with the edge of a slit, with the wave function (3.9) extending accordingly to the sum over all points of both slits. ■

Example 3.6 (Aharonov–Bohm effect). Another example is the Aharonov–Bohm effect [49], where a charged particle is affected by the magnetic potential $\mathbf{A}$ in the absence of any actual magnetic field, $\mathbf{B} = \text{rot } \mathbf{A} = \mathbf{0}$. The presence of $\mathbf{A}$ in the Hamiltonian does affect the classical action in the Hamilton-Jacobi p.d.e. (2.2) and thus the quantum wave constructed in theorem 3.2.

Consider for instance the two-slit set-up of example 3.5, with the same two-connected manifold $\mathbb{G}^2$ as in [figure 1a](#), but assume now that behind the wall ($x_1 > 0$) there is a magnetic potential $\mathbf{A}(\mathbf{x})$, with gauge $\nabla_M \cdot \mathbf{A} = 0$. The classical Hamiltonian-Jacobi p.d.e. (2.2) is

$$-\frac{\partial \phi_j}{\partial t} = H = \frac{1}{2M} (\nabla \phi_j - Q\mathbf{A})^T (\nabla \phi_j - Q\mathbf{A}) \quad \mathbf{x} = (x^1, x^2, x^3)^T \in \mathbb{G}^3 = \mathbb{R}^3 \setminus \mathbb{W}^3$$

with mass M , particle charge Q , wall with two holes $\mathbb{W}^3 = \{x^1 = 0\} \setminus \mathbb{H}^3$, where $\mathbb{H}^3 = \{\mathbf{x}_1, \mathbf{x}_2\}$ with $\mathbf{x}_{1,2} = (0, \pm 5, 0)^T$, and initial momentum p_0 . The $\mathbb{B} = \{1, 2\}$ -valued least action of example 3.5 becomes

$$\phi_j = \begin{cases} p_0 x^1 - Et & \text{for } x^1 < 0 \\ p_0 r_j - Et + Q \int_{\mathbf{x}_j}^{\mathbf{x}} \mathbf{A} \, d\mathbf{x} & \text{for } x^1 \geq 0 \end{cases} \quad \text{where } E = \frac{p_0^2}{2M}.$$

Note that the integral above is path-independent since $\mathbf{A}$ is a gradient field (by hypothesis, $\text{rot } \mathbf{A} = \mathbf{0}$). The vanishing magnetic field $\mathbf{B}$ implies that the classical particle path and hence the classical density in (3.8) are identical to those of the two-slit experiment in [figure 1](#).

Theorem 3.2 now yields the (un-normalized) wave function (3.3),

$$\psi = \sum_{j \in \mathbb{B}} \sqrt{\rho_j} e^{\frac{i}{\hbar} \phi_j} = e^{-\frac{i}{\hbar} Et} \begin{cases} e^{\frac{i}{\hbar} p_0 x^1} & \text{for } x^1 < 0 \\ \frac{1}{r_1} e^{\frac{i}{\hbar} (p_0 r_1 + Q \int_{\mathbf{x}_1}^{\mathbf{x}} \mathbf{A} \, d\mathbf{x})} + \frac{1}{r_2} e^{\frac{i}{\hbar} (p_0 r_2 + Q \int_{\mathbf{x}_2}^{\mathbf{x}} \mathbf{A} \, d\mathbf{x})} & \text{for } x^1 \geq 0. \end{cases}$$

Hence theorem 3.2 provides a classically-based explanation of the Aharonov–Bohm phase shift [49], which is added in the Fraunhofer wave (3.9). ■

Example 3.7 (Particle in a box). Consider the Hamilton-Jacobi p.d.e. (2.2) of a particle in a box of width L in [figure 2](#)

$$-\frac{\partial \phi}{\partial t} = H = \frac{1}{2M} \nabla \phi^2 \quad 0 \leq \frac{x}{L} \leq 1$$

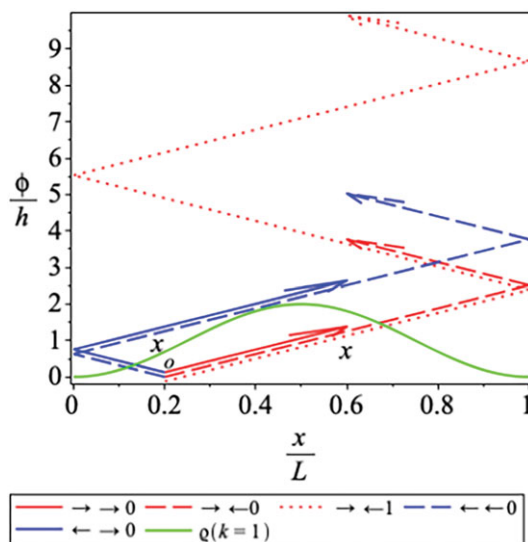


Figure 2. Multipaths from $x_o/L = 0, 2$ to $x/L = 0, 6$ in a box, and the resulting wave.

with position x , initial position x_o , constant mass M , and unknown momentum p . The $\mathbb{B} = \{\{\rightarrow, \leftarrow\} \times \{\rightarrow, \leftarrow\} \times p \in \mathbb{R}_+ \times k \in \mathbb{N}\}$ - valued action of theorem 2.4

$$\phi_j = 2kLp - Et + \begin{cases} p(x - x_o) & \text{for } j = \rightarrow \rightarrow k \\ p(2L - (x + x_o)) & \text{for } j = \rightarrow \leftarrow k \\ p(2L - (x - x_o)) & \text{for } j = \leftarrow \leftarrow k \\ p(x + x_o) & \text{for } j = \leftarrow \rightarrow k \end{cases} \quad E = \frac{p^2}{2M}$$

is illustrated in figure 2. The paths are augmented with $2kL$ periodic path elements since each wall reflection leads to an additional action branch. From lemma 3.4, this implies the quantization $2\frac{Lp}{\hbar} = 2\pi k$, $k \in \mathbb{N}$, yielding $p = \pi k \frac{\hbar}{L}$ so that the number of solutions reduces to $\mathbb{B} = \{\{\rightarrow, \leftarrow\} \times \{\rightarrow, \leftarrow\} \times k \in \mathbb{N}\}$. Using $\Delta_M \phi_k = 0$ outside the edges, the classical density (2.6) $\sqrt{\rho_k} = \sqrt{2/L} \cdot 1/2i$ is constant. Thus, letting $E_k = \hbar^2 \pi^2 k^2 / 2ML^2$, the normalized wave function (3.3) of theorem 3.2 is

$$\psi = \sum_{k \in \mathbb{N}} \sqrt{\rho_k} \sum_{\{\rightarrow, \leftarrow\} \times \{\rightarrow, \leftarrow\}} e^{\frac{i}{\hbar} \phi_j} = \sqrt{\frac{2}{L}} \sum_{k \in \mathbb{N}^*} e^{-\frac{i}{\hbar} E_k t} \sin \frac{\pi k x_o}{L} \sin \frac{\pi k x}{L}.$$

Figure 2 illustrates for $k=1$ the 4 classical “billiard” paths. In addition, one periodic extended path $\rightarrow \rightarrow 1$ is shown. Measuring the particle energy E_k corresponds to selecting a branch k in theorem 2.4. From lemma 3.3, this is associated to a wave collapse. The resulting probability density $\varrho = \psi \psi^\dagger$ is also shown for $k=1$. The novelty is that this well-known result [26,48] is derived just from the constrained, multi-valued action of theorem 2.4. ■

Example 3.8 (Tunnelling). Consider the barrier problem with the Hamilton-Jacobi p.d.e. (2.2),

$$-\frac{\partial \phi}{\partial t} = H = \frac{1}{2M} \nabla \phi^2 + V \quad \text{where } V(x) = \begin{cases} 0 & \text{for } x < 0 \\ V_\infty > 0 & \text{for } x \geq 0 \end{cases} \quad x \in \mathbb{R} \quad (3.10)$$

where particles with constant mass M are shot with an initial momentum p_o and density ρ_o toward the potential barrier $V(x)$ at a position x . Besides the initial path, for $x < 0$ there is a reflected path R , and for $x \geq 0$ the possibility of a transmitted path T . The $\mathbb{B} = \{\rightarrow, \leftarrow\}$ - valued action of theorem

$$\phi_j = \begin{cases} -\frac{p_o^2}{2M} t + p_o x & \text{for } x < 0, j = \rightarrow \\ -\frac{p_o^2}{2M} t - p_o x & \text{for } x < 0, j = \leftarrow \\ -\frac{p_T^2}{2M} t + p_T x & \text{for } x \geq 0, j = \leftarrow, \rightarrow \end{cases}$$

where $p_T = \sqrt{p_o^2 - 2M V_\infty}$ is real for $p_o^2/2M \geq V_\infty$ and imaginary for $p_o^2/2M < V_\infty$, i.e. a complex action exists in both cases. The action is real except for $x > 0$ and $p_o^2/2M < V_\infty$ where it is imaginary.

The position dynamics (2.3), (2.4) of (3.10) is linear for $x > 0$. Thus, superposition of the two imaginary path solutions $M\dot{x}_{1,2} = \pm p_T$ yields the real position solution $x = x_1 + x_2$, with real velocity $\dot{x} = 0$. The conjugate imaginary actions and momenta are well defined solutions of the Hamilton-Jacobi equation (2.2), and correspond by superposition to a real position with zero velocity.

Since $\Delta_M \phi_j = 0$, the classical density (2.6) is constant for $x \neq 0$. At $x = 0$, the classical continuity equation (1.4) and force equilibrium imply

$$\frac{p_o}{M} \rho_o = \frac{p_o}{M} \rho_T + \frac{p_T}{M} \rho_R, \quad (\rho_T + 2\rho_R) \left(\frac{p_o}{M}\right)^2 = (\rho_o + \rho_R) \left(\frac{p_o}{M}\right)^2 = \left(\left(\frac{p_o}{M}\right)^2 + \left(\frac{p_T}{M}\right)^2\right) \rho_T$$

The normalized wave function (3.3) of theorem 3.2 or lemma 3.1 for the complex case is hence

$$\psi = \sum_{j \in \mathbb{B}} \sqrt{\rho_j} e^{\frac{i}{\hbar} \phi_j} = \begin{cases} \left(\sqrt{\rho_o} e^{\frac{i}{\hbar} p_o x} + \sqrt{\rho_R} e^{-\frac{i}{\hbar} p_o x} \right) e^{-\frac{i}{\hbar} \frac{p_o^2}{2M} t} & \text{for } x \leq 0 \\ \sqrt{\rho_T} e^{\frac{i}{\hbar} p_T x - \frac{i}{\hbar} \frac{p_T^2}{2M} t} & \text{for } x > 0 \end{cases}.$$

Again, the novelty is that this well-known wave function [48,50,51] is derived just from the classical multi-valued action and density of theorem 2.4, covering the “classically prohibited” area.

The next examples use Hermite polynomials $H_k(z) = (2z - \frac{d}{dz})^k \cdot 1, k \geq 0$ which yield basis functions orthonormal with respect to the measure e^{-z^2} [48]

$$\psi_{k_n}(x^n) = \sqrt{\frac{M\omega}{\pi \hbar}} \frac{1}{\sqrt{2^{k_n} k_n!}} H_{k_n}(z^n) e^{-1/2(z^n)^2} \quad \text{with } z^n = x^n \sqrt{\frac{M\omega}{\hbar}}. \quad (3.11)$$

■

Example 3.9 (Harmonic oscillator). Consider the Hamilton-Jacobi p.d.e. (2.2) of theorem 2.4

$$-\frac{\partial \phi}{\partial t} = H = \frac{1}{2} \left(\frac{\nabla \phi^T \nabla \phi}{M} + M\omega^2 \mathbf{x}^T \mathbf{x} \right) \quad \mathbf{x} = (x^1, \dots, x^N) \in \mathbb{R}^N, \quad t \geq 0$$

with constant angular frequency ω and mass M , Cartesian position $\mathbf{x}$ and given initial position $\mathbf{x}_0$. The single-valued real least action of theorem 2.4 is

$$\phi = \frac{M\omega}{2} \left(\cot \omega t (\mathbf{x}^T \mathbf{x} + \mathbf{x}_0^T \mathbf{x}_0) - \frac{2}{\sin \omega t} \mathbf{x}^T \mathbf{x}_0 \right) \quad (3.12)$$

which solves the Hamilton-Jacobi p.d.e. [3], with $\phi(\mathbf{x} = \mathbf{x}_0, \omega t = 2k\pi) = 0$ for $k \in \mathbb{Z}$ from L'Hôpital's rule. Using $\Delta_M \phi = N\omega \cot \omega t$, the classical density (2.6) can be computed as $\sqrt{\rho}(\mathbf{x}, t) = \sqrt{M\omega/2i\pi\hbar \sin \omega t}^N$.

The normalized wave function (3.3) of theorem 3.2 is

$$\psi = \sqrt{\rho} e^{\frac{i}{\hbar} \phi} = \sum_{k \in \mathbb{N}}^{\forall k_1 + \dots + k_N = k} e^{-\frac{i}{\hbar} E_k t} \prod_{n=1}^N \psi_{k_n}(x^n) \psi_{k_n}(x_0^n) \quad (3.13)$$

where we used a Taylor series expansion and constants $c_{k_1, \dots, k_N}$, as detailed in [3]. This leads to the eigenvalues $E_k = \hbar \omega(k + \frac{N}{2})$ and wave eigenfunctions (3.11).

The computation above exploits many elements of the Feynman path integral [3], but there is no intrinsic process noise added to the classical path. Feynman [3] showed that no process noise is needed in the particular case of a single-valued quadratic action in $\mathbf{x} \in \mathbb{R}^N$, but did not extend this result to a non-quadratic multi-valued action in a constrained manifold, as is the case in all other examples of this paper. ■

The next example uses quaternion coordinates $\pm \mathbf{q}(\mathbf{x}) = \pm(q^1, \dots, q^4)$, which relate to Cartesian coordinates $\mathbf{x} = (x^1, x^2, x^3)$ [16] as

$$\left. \begin{aligned} x^1 &= 2q^1 q^3 + 2q^2 q^4 \\ x^2 &= -2q^1 q^2 + 2q^3 q^4 \\ x^3 &= (q^1)^2 - (q^2)^2 - (q^3)^2 + (q^4)^2 \end{aligned} \right\} \quad (3.14)$$

For the 2-dimensional case, i.e. $q_3, q_4 = 0$, the above is a complex square root with $q^1 = \pm \sqrt{\frac{1}{2}(\sqrt{(x^2)^2 + (x^3)^2} + x^3)}$, $q^2 = \pm \text{sign}(x^2) \sqrt{\frac{1}{2}(\sqrt{(x^2)^2 + (x^3)^2} - x^3)}$. The kinetic energy can be transformed from Cartesian to quaternion coordinates as $M \dot{\mathbf{x}}^T \dot{\mathbf{x}} = M \dot{\mathbf{q}}^T \frac{\partial \mathbf{x}}{\partial \mathbf{q}} \frac{\partial \mathbf{x}}{\partial \mathbf{q}}^T \dot{\mathbf{q}} = 4M \dot{\mathbf{q}}^T \mathbf{q} \dot{\mathbf{q}}$. Using these coordinates, we now show how the action and wave of an electron around a proton Coulomb field derive from the harmonic oscillator, yielding the basic model of a hydrogen atom. ■

Example 3.10 (Coulomb or gravity potential). Consider a particle in figure 3a with the Hamilton-Jacobi p.d.e. (2.2) of theorem 2.4

$$\left. \begin{aligned} -\frac{\partial \phi}{\partial t} &= H = \frac{1}{2M} \frac{\partial \phi}{\partial \mathbf{x}}^T \frac{\partial \phi}{\partial \mathbf{x}} - \frac{G}{r} & \mathbf{x} &= (x^1, \dots, x^3) \in \mathbb{R}^3, \quad t \geq 0 \\ &= \frac{1}{r} \left(\frac{1}{2 \cdot 4M} \frac{\partial \phi}{\partial \mathbf{q}}^T \frac{\partial \phi}{\partial \mathbf{q}} - G \right) & \mathbf{q} &= (q^1, \dots, q^4) \in \mathbb{R}^4, \quad t \geq 0 \\ \text{and} \quad -\frac{\partial \phi}{\partial t'} &= \frac{1}{2 \cdot 4M} \frac{\partial \phi}{\partial \mathbf{q}}^T \frac{\partial \phi}{\partial \mathbf{q}} - G & t' &= \int_0^t \frac{dt}{r} \quad t = \int_0^{t'} r dt' \end{aligned} \right\} \quad (3.15)$$

with the two-valued quaternion $\pm \mathbf{q}(\mathbf{x})$ of (3.14) initialized at $\pm \mathbf{q}_0$, radius $r = \mathbf{q}^T \mathbf{q} = \sqrt{\mathbf{x}^T \mathbf{x}}$, and constant particle mass M . The constant G corresponds to the Newtonian constant of gravitation scaled with the constant mass of the particle and the mass of the singularity, or equivalently to the Coulomb constant scaled with the constant charge of the particle and the charge of the singularity.

The singularity at the origin is a branch point with a $\mathbb{J} = \{\mathbf{q}_0 \in \mathbb{R}^4\} \times \{\{\uparrow, \downarrow\} \times \omega \in \mathbb{R}_+^* \times k \in \mathbb{N}\}$ - valued real least action [4,5] of theorem 2.4,

$$\phi_j = \frac{4M\omega}{2} \left((\mathbf{q}^T \mathbf{q} + \mathbf{q}_0^T \mathbf{q}_0) \cot(\omega t') \mp \frac{2}{\sin(\omega t')} \mathbf{q}^T \mathbf{q}_0 \right) + \frac{4M}{2} \omega^2 t + G \left(t' + \frac{2\pi k}{\omega} \right)$$

which solves (3.15) with $\phi_j(\mathbf{q} = \mathbf{q}_0, \omega t = 2k\pi) = 0$ for $k \in \mathbb{Z}$ from L'Hôpital's rule, and the turn rate $\omega \in \mathbb{R}_+^*$. The two-valued quaternions $\pm \mathbf{q}(\mathbf{x})$ correspond in figure 3a to the right-turning $\downarrow$ and left-turning $\uparrow$ Kepler orbits [52]. The orbits are augmented with $G \frac{2\pi k}{\omega}$ periodic orbit elements, since each additional full rotation 2π leads to an additional action branch. From lemma 3.4, ω is quantized as $\frac{G}{\hbar} \frac{2\pi}{\omega} = 2\pi k$, $k \in \mathbb{N}^*$, yielding $\omega = G/\hbar k$, so that the number of action solutions reduces to $\mathbb{J} = \{\mathbf{q}_0 \in \mathbb{R}^4\} \times \{\{\uparrow, \downarrow\} \times k \in \mathbb{N}^*\}$.

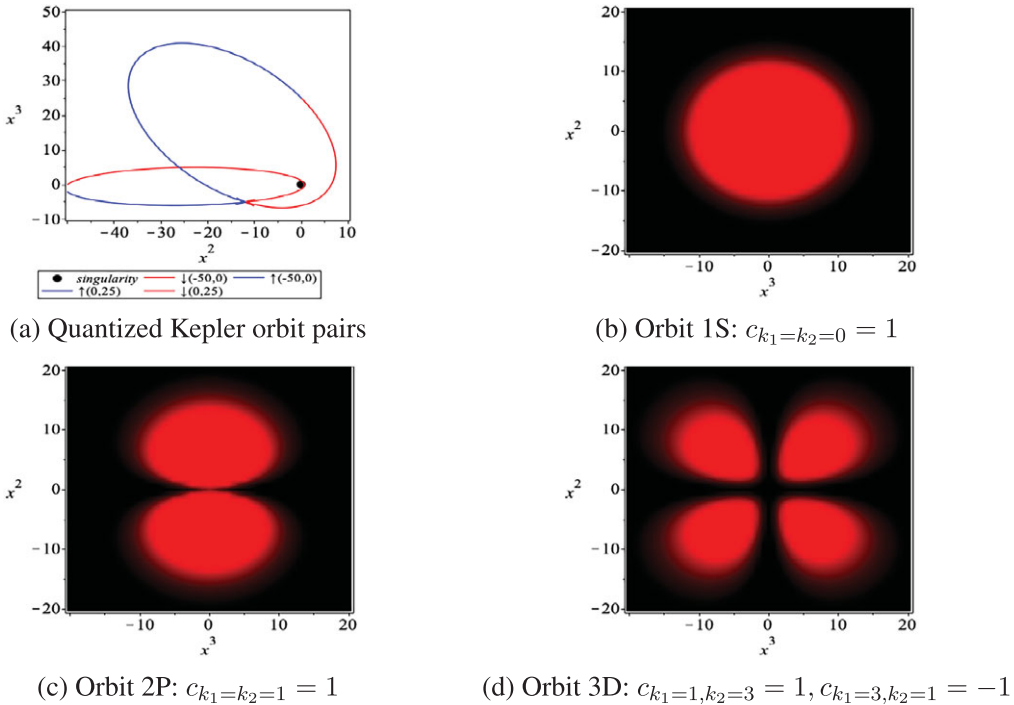


Figure 3. Quantized Kepler orbits and hydrogen orbitals.

For a given $\mathbf{q}_0$, the classical density (2.6) can be computed from $\Delta_M \phi_j = 4\omega \cot \omega t'$ as $\sqrt{\rho_j} = \sqrt{M\omega/2i\pi\hbar \sin \omega t'}$. Using (3.13) from example 3.9, the normalized wave function (3.3) of theorem 3.2 is

$$\begin{aligned} \psi(\mathbf{q}, \mathbf{q}_0, t) &= \sum_{j \in \mathbb{J}} \sqrt{\rho_j} e^{\frac{i}{\hbar} \phi_j} = \sum_{k' \in \mathbb{N}}^{\forall k_1 + \dots + k_4 = k'} c_{k_1..k_4} e^{\frac{i}{\hbar} (G - E_{k'}) t' + \frac{i}{\hbar} E_k t} \prod_{n=1}^4 \psi_{k_n}(q^n) \psi_{k_n}(q_0^n) \\ &= \sum_{k \in \mathbb{N}^*}^{\forall k_1 + \dots + k_4 = 2k-2} c_{k_1..k_4} e^{\frac{i}{\hbar} E_k t} \prod_{n=1}^4 \psi_{k_n}(q^n) \psi_{k_n}(q_0^n) \end{aligned} \quad (3.16)$$

where we used $E_{k'} = \hbar\omega(k' + \frac{4}{2}) = 2\hbar\omega k = G \implies E_k = 2M\omega^2 = M/2 (\frac{G}{\hbar k})^2$ to remove the dependence on t' , and constants $c_{k_1..k_4}$. Note that $k' + 2 = 2k$ is even, due to the symmetry $H_{k_n}(-q^n) = (-1)^{k_n} H_{k_n}(q^n)$ and the multipaths $\pm q^n(\mathbf{x})$ of the wave eigenfunctions $\psi_{k_n}(q^n)$ of (3.11).

Performing with (3.11) a Hilbert space decomposition [53] of the initial classical density distribution $\sqrt{\rho_j}(\mathbf{x}_0, 0) = \sum_{k \in \mathbb{N}^*}^{\forall k_1 + \dots + k_4 = 2k-2} c_{k_1..k_N} \prod_{n=1}^N \psi_{k_n}(x_0^n)$, the wave (3.13) can be rewritten

$$\psi(\mathbf{q}, t) = \sum_{k \in \mathbb{N}^*}^{\forall k_1 + \dots + k_4 = 2k-2} c_{k_1..k_4} e^{\frac{i}{\hbar} E_k t} \prod_{n=1}^4 \psi_{k_n}(q^n).$$

While computed just from the $\mathbb{J} = \mathbf{q}_0 \in \mathbb{R}^4 \times \{\{\uparrow, \downarrow\} \times k \in \mathbb{N}^*\}$ classical counter-rotating Kepler orbits in figure 3a, this result matches the 3-dimensional Coulomb wave in spherical coordinates [4,48,54].

Figures 3a to 3d illustrate the classical Kepler orbits and quantum probability density (3.5, 3.16) in Cartesian coordinates x^2, x^3 (3.14) and time t for $\sqrt{\frac{4M\omega}{\hbar}} = 1, q^3 = 0, q^4 = 0, r = (q^1)^2 + (q^2)^2$. Note that different eigenfunctions of the same degenerate eigenvalue can be superimposed to generate new eigenfunctions of the same degenerate eigenvalue. Also, the superposition principle applies

both to the Coulomb wave (3.16), and to the Kepler orbits, whose position dynamics (3.15) is linear in $\mathbf{q}, t'$ coordinates.

- figure 3a shows two Kepler orbit pairs with starting position $(q_o^1, q_o^2) = (-5, -5)$ or $(0, 5)$, which is in Cartesian coordinates $(x_o^2, x_o^3) = (-50, 0)$ or $(0, 25)$, and end position $(q^1, q^2) = (3, 2)$, which is in Cartesian coordinates $(x^2, x^3) = (-12, -5)$. The two counter rotation solutions correspond to $\pm \mathbf{q}$.
- figure 3b shows the quantum eigendensity $\varrho = \psi \psi^\dagger = \frac{1}{\sqrt{\pi} 2^0 0!} (H_0(q^1) H_0(q^2))^2 e^{-r/2} = (1/\sqrt{\pi}) e^{-r/2}$ for $k=1, c_{k_1=k_2=0}=1$ of (3.16), called orbit 1S.
- figure 3c shows the quantum eigendensity $\varrho = \psi \psi^\dagger = \frac{1}{\sqrt{\pi} 2^1 1!} (H_1(q^1) H_1(q^2))^2 e^{-r/2} = 1/2 \sqrt{\pi} (4q^1 q^2)^2 e^{-r/2} = 2/\sqrt{\pi} (x^2)^2 e^{-r/2}$ for $k=2, c_{k_1=k_2=1}=1$ of (3.16), called orbit 2P. The 2P orbit can be superimposed to the 2S orbit, which is the 2P orbit rotated by 90° with the same eigenvalue.
- figure 3d shows the quantum eigendensity $\varrho = \psi \psi^\dagger = \frac{1}{\sqrt{\pi} (2^1 1!) (2^3 3!)} (H_1(q^1) H_3(q^2) - H_3(q^1) H_1(q^2))^2 e^{-r/2}$ for $k=3, c_{k_1=1, k_2=3}=1, c_{k_1=3, k_2=1}=-1$ of (3.16), called orbit 3D.

There is no process noise added to the classical path in contrast to the Feynman path integral [3] in the Duru-Kleinert propagator [4]. Hence the eigenwaves are now derived from the $\mathbb{J} = \{\mathbf{q}_o \in \mathbb{R}^4\} \times \{\{\uparrow, \downarrow\} \times k \in \mathbb{N}^*\}$ determined Kepler orbits [52].

One of the motivations for Bohr's atom model in the early days of quantum mechanics was the instability of previous orbital models, where the magnetic field generated by the electron's circular motion led to continual radiation and thus collapse of the atom. This problem is avoided here, since the effects of the two counter-rotating magnetic fields cancel each other. ■

4. The relativistic case

Lemma 3.1 is actually a general mapping from a quadratic first-order p.d.e. to a linear second-order p.d.e., i.e. it is not restricted to the Hamilton-Jacobi or Schrödinger equations. Specifically, the following three sections will apply lemma 3.1 and theorem 3.2 to

- The Klein-Gordon equation, by replacing the classical action ϕ with a relativistic action.
- The Dirac and Pauli equations, by replacing the classical action ϕ with a unit quaternion action.
- The relativistic Maxwell equation, by replacing the classical action ϕ with a relativistic action with rest mass $M_o = 0$.

In general relativity, the metric tensor $\mathbf{M}'$ is defined by the Einstein field equation (EFE) [55]. The metric has a direct impact on the action and therefore on the wave in (3.2). As such the EFE, which includes gravity, is implicitly covered in the three relativistic cases above. In special relativity, the metric is the Minkowski metric

$$\mathbf{M} = M_o \mathbf{M}' \quad \mathbf{M}' = \text{diag}(RI^2 c^2, -1, -1, -1) \quad (4.1)$$

with constant rest mass M_o , speed of light constant c , and refractive index $RI \leq 1$ (where $RI = 1$ in free space). We use $\mathbf{M}$ for massive particles and $\mathbf{M}'$ for massless particles.

(a) Klein-Gordon equation

The Hamilton-Jacobi p.d.e. (2.2) also applies to general relativity [55,56] when we replace

$$\left. \begin{aligned} \mathbf{x} \rightarrow \bar{\mathbf{x}} &= \begin{pmatrix} t \\ \mathbf{x} \end{pmatrix}, \quad \mathbf{A} \rightarrow \bar{\mathbf{A}} = \begin{pmatrix} V_E(\bar{\mathbf{x}}) \\ \mathbf{A}(\bar{\mathbf{x}}) \end{pmatrix}, \quad \mathbf{p} = \frac{\partial \phi}{\partial \mathbf{x}} \rightarrow \bar{\mathbf{p}} = \frac{\partial \phi}{\partial \bar{\mathbf{x}}} = \begin{pmatrix} E(\bar{\mathbf{x}}) \\ \mathbf{p}(\bar{\mathbf{x}}) \end{pmatrix} \\ \text{and} \quad t \rightarrow \tau &= \int \sqrt{\frac{d\bar{\mathbf{x}}^T}{dt} \mathbf{M}'(\bar{\mathbf{x}}) \frac{d\bar{\mathbf{x}}}{dt} \frac{dt}{c}} \geq 0 \end{aligned} \right\} \quad (4.2)$$

with classical momentum $\mathbf{p}$, relativistic total energy E , electrostatic potential V_E and vector potential $\mathbf{A}$, charge Q , and proper time τ path integral, measured by a clock attached to the particle. In proper time, i.e. in a coordinate frame attached to the particle, the energy of a particle is always the constant rest energy $E_0 = M_0 c^2$. Hence, the action in theorem 2.4 and the wave in theorem 3.2 transform as

$$\phi_j(\mathbf{x}, t) \rightarrow \Phi_j(\bar{\mathbf{x}}) \quad \psi_j(\mathbf{x}, t) \rightarrow \psi_j(\bar{\mathbf{x}}) \quad \bar{\mathbf{x}} \in \mathbb{G}^4.$$

We thus have the relativistic covariant Hamilton-Jacobi p.d.e. for a single particle, otherwise known as the relativistic energy momentum relation [55,57], for $j \in \mathbb{J}$

$$\left. \begin{aligned} -\frac{\partial \phi_j}{\partial \tau} = 0 = H &= \frac{1}{2} \frac{d\bar{\mathbf{x}}^T}{d\tau} \mathbf{M} \frac{d\bar{\mathbf{x}}}{d\tau} - \frac{E_0}{2} \\ \text{and} \quad \frac{d\nabla \phi_j}{d\tau} + \frac{\partial H}{\partial \bar{\mathbf{x}}} &= \sum_{g \in \mathbb{G}} \frac{\partial f_g}{\partial \bar{\mathbf{x}}} \lambda_g, \quad \mathbf{M} \frac{d\bar{\mathbf{x}}}{d\tau} = \nabla \phi_j - Q \bar{\mathbf{A}} \end{aligned} \right\} \quad (4.3)$$

Note that in standard derivations of the relativistic quantum equations [29,30,32,48], the Hamiltonian is set to E and not to H in (4.3). This prevents a direct generalization of the Schrödinger equation (1.6) to the relativistic case, as we do here and in the next subsections. The relativistic density is given by the relativistic continuity equation

$$0 = \frac{\partial}{\partial \tau} \rho_j + \nabla \mathbf{M} \cdot (\rho_j \dot{\bar{\mathbf{x}}}) = \frac{d\rho_j}{d\tau} + \rho_j \nabla \mathbf{M} \cdot \dot{\bar{\mathbf{x}}} \quad (4.4)$$

whose solution always remains positive. Using lemma 3.1 yields the familiar Klein-Gordon equation [29,30,56]

$$0 = \left[\left(\frac{\hbar}{i} \nabla \mathbf{M} - Q \bar{\mathbf{A}} \right) \cdot \mathbf{M}^{-1} \left(\frac{\hbar}{i} \nabla - Q \bar{\mathbf{A}} \right) - E_0 \right] \psi(\bar{\mathbf{x}}). \quad (4.5)$$

Hence the Klein-Gordon equation is a special case of theorems 2.4 and 3.2 for unconstrained $\bar{\mathbf{x}} \in \mathbb{R}^4$. Note that theorem 3.2 can be seen as an extension of the non-relativistic zig-zag Feynman path integral (3.7) of [2] to the relativistic case, respecting the speed of light limit.

Importantly, the proper density (4.4) is defined with respect to a proper 4-dimensional volume element, which does not Lorentz contract [57,58], by contrast to a three-dimensional volume. Hence the quantum density matrix $\varrho = \psi \psi^\dagger$ of (3.5) is the coordinate invariant positive probability to find a particle in a four-dimensional volume element. Replacing t by τ in (3.6), ϱ normalized at $\tau = \tau_0$ remains normalized $\forall \tau \geq \tau_0$.

(b) Dirac and Pauli equations

Quaternion rotation of objects [16] is well defined in geometry, and is used in an identical form in classical, relativistic, and quantum physics. This section defines accordingly a quaternion-based rotation action, and shows how the Dirac and Pauli equations can be derived from this classical

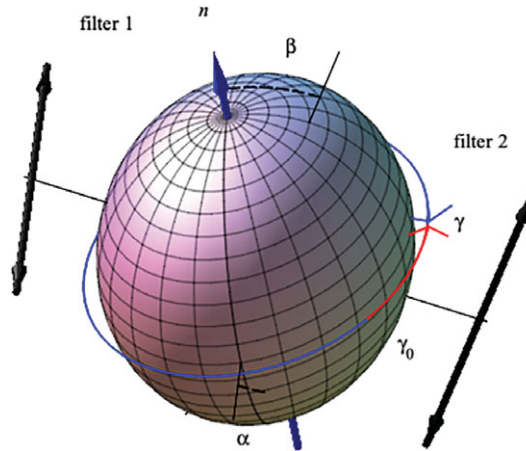


Figure 4. Two classical counter rotations.

quaternion action. We use standard Pauli spin [31] and Dirac matrices [22,32],

$$\Sigma = (\sigma^1, \sigma^2, \sigma^3) \quad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4.6)$$

and

$$\Gamma = (\gamma^0, \gamma^1, \gamma^2, \gamma^3) \quad \gamma^0 = \begin{pmatrix} \mathbf{I} & 0 \\ 0 & -\mathbf{I} \end{pmatrix} \quad \gamma^n = \begin{pmatrix} 0 & \sigma^n \\ \sigma^n & 0 \end{pmatrix} \quad \text{for } n = 1, 2, 3. \quad (4.7)$$

In the non-relativistic case, we replace the scalar action in theorem 2.4 with a 2×2 pure imaginary quaternion action, which implies a unit quaternion wave in lemma 3.1,

$$\phi_j \rightarrow \phi_j = \hbar \Sigma \cdot \mathbf{n} \frac{s\gamma}{2} \in \mathbb{H} \quad (4.8)$$

and

$$\psi_j \rightarrow \psi_j = \mathbf{I} \cos \frac{s\gamma}{2} + i \Sigma \cdot \mathbf{n} \sin \frac{s\gamma}{2} = e^{i \Sigma \cdot \mathbf{n} \frac{s\gamma}{2}} \in \mathbb{H}_1 \quad (4.9)$$

with the roll angle rotation $-\pi \leq s\gamma \leq \pi$ in figure 4 around the unit direction $\mathbf{n} = (n^1, n^2, n^3)^T = (\sin \beta \cos \alpha, \sin \beta \sin \alpha, \cos \beta)^T$, with Euler yaw $-\pi \leq \alpha \leq \pi$ and pitch $0 \leq \beta \leq \pi$. The particles are rotationally symmetric of order $2\pi/s$ [59], with $s \in \mathbb{N}^*$. Fermions and bosons correspond to $s = 1$ and $s = 2$. The 2×2 unit quaternion (4.9) can be modally decomposed as

$$\Sigma \cdot \mathbf{n} = \begin{pmatrix} \cos \beta & e^{-i\alpha} \sin \beta \\ e^{i\alpha} \sin \beta & -\cos \beta \end{pmatrix} = \chi^\uparrow (\chi^\uparrow)^\dagger - \chi^\downarrow (\chi^\downarrow)^\dagger \quad (4.10)$$

and

$$\chi^\uparrow = \begin{pmatrix} \cos \frac{\beta}{2} \\ e^{i\alpha} \sin \frac{\beta}{2} \end{pmatrix} \quad \chi^\downarrow = \begin{pmatrix} -e^{-i\alpha} \sin \frac{\beta}{2} \\ \cos \frac{\beta}{2} \end{pmatrix}. \quad (4.11)$$

The orthonormal eigenvectors $\chi^\uparrow(\mathbf{n})$ and $\chi^\downarrow(\mathbf{n})$ of $\Sigma \cdot \mathbf{n}$, called eigenspinors [16], correspond to an aligned $\uparrow$ or anti-aligned $\downarrow$ classical rotation around $\mathbf{n}$ in figure 4 of the Bloch sphere [60]. In contrast to Euler angles, unit quaternions yield a linear motion description and hence allow a simple application of theorems 2.4 and 3.2. Note that translational and rotational dynamics are uncoupled (in the absence of external coupling forces).

In the relativistic case, we replace instead the scalar action in theorem 2.4 with a pure imaginary 4×4 quaternion action, which also implies a unit quaternion wave in lemma 3.1,

$$\phi_j \rightarrow \phi_j = \hbar \Gamma \cdot \bar{\mathbf{p}} \frac{s\gamma}{2} \in \mathbb{H} \quad (4.12)$$

and

$$\psi_j \rightarrow \psi_j = \mathbf{I} \cos \frac{s\gamma}{2} + \frac{i}{\hbar} \Gamma \cdot \bar{\mathbf{p}} \sin \frac{s\gamma}{2} = e^{\frac{i}{\hbar} \Gamma \cdot \bar{\mathbf{p}} \frac{s\gamma}{2}} \in \mathbb{H}_1. \quad (4.13)$$

Both wave quaternions are differentiable for constant $\mathbf{n}$ and $\bar{\mathbf{p}}$. The 4×4 quaternions (4.12) can be modally decomposed as [32,56]

$$c\Gamma(\alpha_p, \beta_p) \cdot \bar{\mathbf{p}} = E_+ \xi^{\uparrow+} \xi^{\uparrow++} + E_+ \xi^{\downarrow+} \xi^{\downarrow++} + E_- \xi^{\uparrow-} \xi^{\uparrow--} + E_- \xi^{\downarrow-} \xi^{\downarrow--} \quad (4.14)$$

where the ambiguity of having four relativistic eigenspinors (4.15) for only two eigen-energies is resolved by using the classical eigenspinors $\chi^{\downarrow\uparrow}(\alpha_p, \beta_p)$ of (4.11) (see figure 4),

$$\left. \begin{aligned} \xi^{\uparrow, \downarrow+} &= \frac{1}{N_+} \begin{pmatrix} \mathbf{I} \\ \Sigma \cdot \mathbf{p} \end{pmatrix} \chi^{\uparrow, \downarrow} & \xi^{\uparrow, \downarrow-} &= \frac{1}{N_+} \begin{pmatrix} \Sigma \cdot \mathbf{p} \\ \mathbf{I} \end{pmatrix} \chi^{\uparrow, \downarrow} \end{aligned} \right\}. \quad (4.15)$$

and

$$E_{\pm} = \pm \sqrt{E_o^2 + \mathbf{p}^T \mathbf{p} c^2} \quad \Delta_{\pm} = E_{\pm} \pm E_o \quad N_{\pm} = \sqrt{1 + \frac{c^2 \mathbf{p}^T \mathbf{p}}{\Delta_{\pm}^2}}$$

The orthonormal eigenvectors $\xi_{\uparrow, \downarrow}^+$, $\xi_{\uparrow, \downarrow}^-$, called eigenspinors [16], correspond to an aligned $\uparrow$ or anti-aligned $\downarrow$ relativistic rotation around $\mathbf{n}$ in figure 4 for a positive or negative eigen-energy $E_{\pm}$. Note for massless particles the 4×1 Dirac spinor (4.15) simplifies into two decoupled 2×1 complex chiral spinors.

The actions and waves (4.8)–(4.13) are quaternions due to the anti-commutation (decoupling) relation

$$\{\sigma^j, \sigma^k\} = \sigma^j \sigma^k + \sigma^k \sigma^j = 2\delta^{jk} \quad (4.16)$$

and

$$\{\gamma^j, \gamma^k\} = \gamma^j \gamma^k + \gamma^k \gamma^j = 2\delta^{jk}. \quad (4.17)$$

Using lemma 3.1, the classical action (4.8) now implies the Pauli equation

$$\left[\frac{\hbar}{i} \frac{\partial}{\partial t} + \frac{1}{2} \Sigma \cdot \left(\frac{\hbar}{i} \nabla - Q\mathbf{A} \right) \cdot \mathbf{M}^{-1} \Sigma \cdot \left(\frac{\hbar}{i} \nabla - Q\mathbf{A} \right) + V \right] \psi = 0. \quad (4.18)$$

The scalar wave function ψ is replaced by a 2×1 complex spinor (4.11).

Still using lemma 3.1, the relativistic action (4.12) now implies the Dirac equation

$$0 = \left[\Gamma \cdot \left(\frac{\hbar}{i} \nabla - Q\bar{\mathbf{A}} \right) + M_o c \right] \psi \quad (4.19)$$

multiplied from the left with the invertible matrix $[\Gamma \cdot (\frac{\hbar}{i} \nabla - Q\bar{\mathbf{A}}) - M_o c] \cdot \mathbf{M}^{-1}$. We use the 4×1 relativistic spinor (4.15). Hence the Pauli or Dirac equation are for $\phi \in \mathbb{R} \rightarrow \phi \in \mathbb{H}$ special cases of theorem 2.4 and theorem 3.2.

(c) Maxwell equation

The action and motion of a photon are given relativistically by (4.3), with zero rest mass $M_o = 0$, zero rest energy $E_o = 0$ and without external forces [55]. We use the relativistic metric $\mathbf{M}'$ of (4.1) instead of the relativistic inertia tensor $\mathbf{M}$. Hence theorem 2.4 and (4.3) yields the relativistic eikonal equation of geometric ray optics [61],

$$0 = H = \frac{1}{2} \nabla \Phi_j^T \mathbf{M}'^{-1} \nabla \Phi_j, \quad (4.20)$$

i.e. a photon moves along a geodesic of the relativistic metric $\mathbf{M}'$ of Einstein's field equation (EFE)

[55]. We now replace in lemma 3.1 the wave of a massive particle with the real relativistic 4×1

vector potential $\bar{\mathbf{A}}'$ of the photon

$$\psi_j \rightarrow \bar{\mathbf{A}}' = \sqrt{\rho_j} e^{\frac{i\phi_j}{\hbar}}$$

where $\sqrt{\rho_j}$ is a 4×1 square root density vector. Now lemma 3.1 yields the familiar homogeneous Maxwell equation [33]

$$\Delta_{\mathbf{M}} \bar{\mathbf{A}}' = \mathbf{0} \quad (4.21)$$

which are four decoupled Laplacians. So far the norm $|\bar{\mathbf{A}}'|$ of the vector potential has remained constant, and hence photons cannot be created nor annihilated in (4.20) or (4.21) by changing the photon density $\sqrt{\rho_j} = \mathbf{0}$ to $\sqrt{\rho_j} \neq \mathbf{0}$, or vice-versa. This is not surprising, since (4.20) is just the geodesic path of a single photon. Photon creation and annihilation can be obtained by replacing (4.21) with the non-homogeneous Maxwell equation

$$\Delta_{\mathbf{M}} \bar{\mathbf{A}} = \mu_0 \bar{\mathbf{J}} \quad (4.22)$$

where μ_0 is the magnetic permeability and $\bar{\mathbf{J}}(\bar{\mathbf{x}})$ is the given relativistic current density. Since Maxwell's homogeneous equation (4.21) is linear, (4.22) can be solved by convolution,

$$\bar{\mathbf{A}}(\bar{\mathbf{x}}) = \mu_0 \int_{\bar{\mathbf{z}} \in \mathbb{G}^N} \bar{\mathbf{A}}'(\bar{\mathbf{x}} - \bar{\mathbf{z}}) \bar{\mathbf{J}}(\bar{\mathbf{z}}) d\bar{\mathbf{z}}$$

where all products are performed element-wise. Thus $\mu_0 \bar{\mathbf{J}}$ changes the norm $|\bar{\mathbf{A}}|$, which implies the creation and annihilation of photons. Hence, with $\psi \in \mathbb{R}$ replaced by $\bar{\mathbf{A}} \in \mathbb{R}^4$, Maxwell's equation is also a special case of theorems 2.4 and 3.2.

(d) Examples with multiple particles

We now turn to examples involving multiple spinning particles, starting with the Einstein-Podolsky-Rosen experiment [62–66]. The fact that the quantum wave in the experiment can be computed using a finite sum of classically-based terms puts its discussion in a different light. We first detail this computation, and then relate it to Bell's Theorem.

Example 4.1 (EPR experiment and entanglement). Consider P spinning particles. Since there is no potential energy, the particles $p \in \mathbb{P} = \{1, \dots, P\}$ are classically decoupled. This implies that the total Hamiltonian and action sum up as $\mathbf{H} = \sum_{p \in \mathbb{P}} \mathbf{H}_p$ and $\phi = \sum_{p \in \mathbb{P}} \phi_p$. We describe the rotation of each individual particle p with a pure imaginary quaternion action (4.8) in the decoupled Hamilton-Jacobi p.d.e. (2.2),

$$-\frac{\partial \phi_p}{\partial t} = \mathbf{H}_p = \frac{1}{2M} \left(\nabla \phi_p^T \nabla \phi_p - \frac{s^2 \hbar^2}{4} \mathbf{I} \right) \quad \phi_p \in \mathbb{H}$$

with the classical eigenspinors $\chi_p^{\uparrow, \downarrow}(\mathbf{n}_p)$ (4.11) of unit rotation direction $\mathbf{n}_p(\alpha_p, \beta_p)$ and roll or spin angle $0 \leq \gamma_p \leq 2\pi/s$ in figure 4. The initial rotation direction is given by the initial classical eigenspinors $\chi_{p0}^{\uparrow, \downarrow}(\mathbf{n}_{p0})$ of the initial unit rotation direction $\mathbf{n}_{p0}(\alpha_{p0}, \beta_{p0})$ of (4.11). The particles have the same moment of inertia M and they are rotationally symmetric of the same order $2\pi/s$, with $s \in \mathbb{N}^*$.

The $\mathbb{J} = \{\gamma_{p0} \in [0, 2\pi/s] + \gamma_p\} \times \{\uparrow, \downarrow\}$ -valued least quaternion action (4.8) of each particle p in theorem 2.4 is

$$\frac{1}{\hbar} \phi_{pj}(\mathbf{n}_p, \gamma_p) = \Sigma \cdot \mathbf{n}_p \begin{cases} \frac{s}{2}(\gamma_p - \gamma_{p0}) & \text{for } \uparrow \\ \frac{s}{2}(\frac{2\pi}{s} - (\gamma_p - \gamma_{p0})) & \text{for } \downarrow \end{cases} \in \mathbb{H} \quad (4.23)$$

where we exploited the anti-commutation relation (4.16) of the Pauli matrices (4.6). While the Hamiltonian $\mathbf{H}_p$ is zero, the action is non-zero and thus the particle can rotate to the left and right in figure 4 with constant spin momentum $\nabla \phi_{pj} = \pm s/2 \hbar \Sigma \cdot \mathbf{n}_p$.

Since $\Delta_M \phi_{pj} = \mathbf{0}$ in (2.6), an initial classical Dirac spinor distribution remains a Dirac density distribution for all $t \geq 0$, i.e. we can define the spinor density as $2i\sqrt{\rho_{pj}^\epsilon} = \chi_p^\epsilon = \chi_{p0}^\epsilon$ with $\epsilon \in \mathbb{E}_p = \{\uparrow, \downarrow\}$.

Thus, from theorem 3.2 and (4.9), the wave spinor (4.11) of particle p ,

$$\psi_p^\epsilon(\mathbf{n}_p, \chi_{p0}^\epsilon) = \sum_{j \in \mathbb{J}} \sqrt{\rho_{pj}^\epsilon} e^{\frac{i}{\hbar} \phi_{pj}} = \int_{\gamma_{p0}=\gamma_p}^{\gamma_p+\frac{2\pi}{s}} \sin \frac{s}{2} (\gamma_p - \gamma_{p0}) d\gamma_{p0} \Sigma \cdot \mathbf{n}_p \chi_{p0}^\epsilon = \Sigma \cdot \mathbf{n}_p \chi_{p0}^\epsilon \quad (4.24)$$

solves the Pauli equation (4.18) with every summand $\sqrt{\rho_{pj}^\epsilon} e^{\frac{i}{\hbar} \phi_{pj}}$. Equation (4.24) computes the wave spinor ψ_p^ϵ

- as the geometric projection of an initial classical spinor χ_{p0}^ϵ on a filter of direction $\mathbf{n}_p(\alpha_p, \beta_p)$. The filter is a branch point of the action (4.23), since it changes the spin direction of the particle from $\mathbf{n}_{p0}$ to $\mathbf{n}_p$.
- as a solution of the eigenspinor equation (4.10) of a pure left or right rotation for $\psi_p^\epsilon = \chi_p^\epsilon = \chi_{p0}^\epsilon$ in free space. A superposition of wave spinors ψ_p^ϵ derives from a classical density ensemble (2.6) of initial χ_{p0}^ϵ .

Finally, aside from initial conditions, the total action $\phi = \sum_{p \in \mathbb{P}} \phi_p(\mathbf{x}_p, t)$ consists of P decoupled pure imaginary quaternion action fields. Hence the corresponding total wave (3.3) consists of P decoupled spinors, and thus may be represented as the tensor product of the individual spinors with $\epsilon \in \mathbb{E} = \{\mathbb{E}_1, \dots, \mathbb{E}_P\}$

$$\psi^\epsilon(\mathbf{n}_p, \chi_{p0}^\epsilon) = \sum_{p \in \mathbb{P}} \psi_1^\epsilon \otimes \dots \otimes \psi_P^\epsilon = \sum_{p \in \mathbb{P}} \Sigma \cdot \mathbf{n}_1 \chi_{10}^\epsilon \otimes \dots \otimes \Sigma \cdot \mathbf{n}_P \chi_{P0}^\epsilon. \quad (4.25)$$

Consider now the Einstein-Podolsky-Rosen (EPR) experiment [62,64–66] with two particles $p=1,2$ (each as in figure 4) which initially have opposite spins in the two ensembles $\mathbb{E} = \{\uparrow\downarrow, \downarrow\uparrow\}$ of probability $p^\epsilon = \frac{1}{2}$. Both ensembles have a classical total spin of 0. Thus the spin of the particles is $\psi_o = \frac{1}{\sqrt{2}}(\psi_o^\uparrow \otimes \psi_o^\downarrow - \psi_o^\downarrow \otimes \psi_o^\uparrow)$ with an unknown initial spin direction $\mathbf{n}_o(\alpha_o, \beta_o)$. Later on, particle 1 is measured behind a filter 1 with angles $\mathbf{n}_1(\alpha_1, \beta_1)$ and at a far distance particle 2 is measured behind a filter 2 with angles $\mathbf{n}_2(\alpha_2, \beta_2)$ in figure 4. With (4.25) this yields

$$\psi = \frac{1}{\sqrt{2}}(\psi_1^\uparrow \otimes \psi_2^\downarrow - \psi_1^\downarrow \otimes \psi_2^\uparrow) \quad \text{where } \psi_p^\uparrow = \Sigma \cdot \mathbf{n}_p \chi_o^\uparrow, \quad \psi_p^\downarrow = \Sigma \cdot \mathbf{n}_p \chi_o^\downarrow. \quad (4.26)$$

The two spin measurements are the geometric projections of an unknown initial spin direction $\chi_o^\uparrow(\alpha_o, \beta_o)$, $\chi_o^\downarrow(\alpha_o, \beta_o)$ on filters of directions $\Sigma \cdot \mathbf{n}_1$ and $\Sigma \cdot \mathbf{n}_2$. The correlation of the two measurements is given by the scalar product of the spinor measurements

$$\langle \psi_1^\uparrow, \psi_2^\downarrow \rangle = \frac{1}{2}(\psi_1^{\uparrow\dagger} \psi_2^\downarrow + \psi_2^{\downarrow\dagger} \psi_1^\uparrow) = \langle \Sigma \cdot \mathbf{n}_1 \chi_o^\uparrow, \Sigma \cdot \mathbf{n}_2 \chi_o^\downarrow \rangle = -\mathbf{n}_1^T \mathbf{n}_2 \quad (4.27)$$

This expression is consistent with experimental results [65]. The second ensemble $\psi_1^\downarrow \otimes \psi_2^\uparrow$ has the same correlation.

For a free particle, the Hamilton-Jacobi formulation of theorem 2.4, as well as Feynman's momentum propagator formulation in [3], determine a constant classical linear or angular momentum $\chi_o^\uparrow, \chi_o^\downarrow$ over time. Equation (4.27) thus allows a classical interpretation that the spins in the distant detectors are correlated through the common initial classical spinors $\chi_o^\uparrow, \chi_o^\downarrow$. ■

Let us put this result in historical perspective. Bell [63] derived the correlation of two classical spinning particles as the integral over a probability density function $\rho(\lambda) \in \mathbb{R}$ of a hypothetical

hidden parameter $\lambda \in \mathbb{R}$,

$$P(\mathbf{n}_1, \mathbf{n}_2) = \int \rho(\lambda) A_1(\mathbf{n}_1, \lambda) A_2(\mathbf{n}_2, \lambda) d\lambda \quad \text{with} \quad \int \rho(\lambda) d\lambda = 1. \quad (4.28)$$

The spins are measured behind unit filters $p = 1, 2$ of direction $\mathbf{n}_p(\alpha_p, \beta_p)$ with the quantized detectors $A_p(\mathbf{n}_p, \lambda) = \pm 1$, or equivalently

$$A_p(\mathbf{n}_p, \lambda) = \chi_p^\uparrow, \chi_p^\downarrow \quad \text{where} \quad \chi_p^\uparrow = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \quad \text{and} \quad \chi_p^\downarrow = \begin{pmatrix} 0 \\ 1 \end{pmatrix} \quad (4.29)$$

leading to Bell's inequality [63]

$$|P(\mathbf{n}_1, \mathbf{n}_2) - P(\mathbf{n}_1, \mathbf{n}_3)| \leq 1 + P(\mathbf{n}_2, \mathbf{n}_3) \quad (4.30)$$

with a third spin direction $p = 3$. The correctness of (4.27) and incorrectness of (4.30) were experimentally demonstrated in [64,65]. Hence it was argued that an exact classical or relativistic derivation of (4.27) is impossible since it violates (4.30).

Spinors or eigenspinors (4.11) on a unit Bloch sphere are exact quaternion descriptions of both classical rotations χ_p^ϵ and quantum rotations ψ_p^ϵ . Although the *classical Bell detector* (4.29) depends on the local filter direction $\mathbf{n}_p$, the quantization is binary and corresponds to relative Euler angles $\beta = 2k\pi, k \in \mathbb{Z}$ on the unit Bloch sphere. Generalizing the quantization of this classical binary Bell detector (4.29) to the full unit Bloch sphere allows to use the *classical spinor detector pair* (4.26),

$$A_1 = \Sigma \cdot \mathbf{n}_1 \chi_0^\downarrow \quad \text{and} \quad A_2 = \Sigma \cdot \mathbf{n}_2 \chi_0^\uparrow. \quad (4.31)$$

Using in Bell's hidden parameter equation (4.28) the general classical spinor detector pair (4.31) in the directions of the filters, instead of the classical binary Bell detector (4.29), directly leads to (4.27) rather than (4.30). Hence the classical correlation of two spinning particles is exactly described by (4.27), whereas Bell's inequality (4.30) is classically only applicable at a relative angle $\beta = 2k\pi, k \in \mathbb{Z}$ on the Bloch sphere. For each particle, what is actually measured of course is just the up/down spin direction behind the filter, not the location on the Bloch sphere. However, the measurement result does depend on the filter, and the classical spinor detector is a description of the *combination* of the filter direction and the spin detector for the particle.

The next example analyzes relativistic spin similarly.

Example 4.2 (Positron and electron creation in quantum electrodynamics (QED)). Consider P relativistically spinning particles. The total Hamiltonian and action sums up as $\mathbf{H} = \sum_{p \in \mathbb{P}} \mathbf{H}_p, \phi = \sum_{p \in \mathbb{P}} \phi_p$. We describe the rotation of each individual particle p with a pure imaginary quaternion action (4.12) in the decoupled Hamilton-Jacobi p.d.e. (2.2),

$$-\frac{\partial \phi_p}{\partial \tau} = \mathbf{H}_p = \frac{1}{2} \left(\nabla \phi_p \mathbf{M}^{-1} \nabla \phi_p - \frac{s^2 \hbar^2 E_o^2}{4} \mathbf{I} \right) = \mathbf{0} \quad \phi_p \in \mathbb{H}$$

with Minkowski metric $\mathbf{M}$ (4.1), rest energy E_o , constant relativistic linear momentum $\bar{\mathbf{p}}_p$ and the relativistic eigenspinors $\xi_p^{\pm\uparrow, \downarrow}(\mathbf{n}_p, \bar{\mathbf{p}}_p)$ (4.15) of unit rotation direction $\mathbf{n}_p(\alpha_p, \beta_p)$ and roll or spin angle $0 \leq \gamma_p \leq 2\pi/s$ in figure 4. The initial rotation direction is given by the initial relativistic eigenspinors $\chi_{p0}^{\pm\uparrow, \downarrow}$. The particles are rotationally symmetric of order $2\pi/s$, with $s \in \mathbb{N}^*$. Electrons and positrons correspond to $s = 1$.

The $\mathbb{J} = \{\gamma_{p0} \in [0, 2\pi/s] + \gamma_p\} \times \{\uparrow, \downarrow\}$ - valued quaternion least action (4.17) of each particle p in theorem 2.4 is

$$\frac{1}{\hbar} \phi_{pj}(\mathbf{n}_p, \bar{\mathbf{p}}_p, \gamma_p) = \Gamma \cdot \bar{\mathbf{p}}_p \begin{cases} \frac{s}{2}(\gamma_p - \gamma_{p0}) & \text{for } \uparrow \\ \frac{s}{2}(\frac{2\pi}{s} - (\gamma_p - \gamma_{p0})) & \text{for } \downarrow \end{cases} \in \mathbb{H} \quad (4.32)$$

where we exploited the commutation properties of the Dirac matrices (4.7). The actions correspond to the left and right rotations in figure 4.

Since $\Delta_M \phi_{pj} = \mathbf{0}$ in (2.6), an initial classical Dirac spinor distribution remains a Dirac density distribution for all $t \geq 0$, i.e. we can define the spinor density as $2i\sqrt{\rho_{pj}}^{\pm\epsilon} = \xi_p^{\pm\epsilon} = \xi_{p0}^{\pm\epsilon}$ with $\epsilon \in \mathbb{E}_p = \{\uparrow, \downarrow\}$. Thus, from theorem 3.2 and (4.10), the relativistic wave spinor (4.15) of particle p ,

$$\psi_p^{\pm\epsilon}(\mathbf{n}_p, \bar{\mathbf{p}}_p, \xi_{p0}^{\pm\epsilon}) = \sum_{j \in \mathbb{J}} \sqrt{\rho_{pj}} e^{i\hbar\phi_{pj}} = \int_{\gamma_{p0}=\gamma_p}^{\gamma_p+\frac{2\pi}{s}} \sin \frac{s}{2} (\gamma_p - \gamma_{p0}) d\gamma_{p0} \Gamma \cdot \bar{\mathbf{p}}_p \xi_{p0}^{\pm\epsilon} = \Gamma \cdot \bar{\mathbf{p}}_p \xi_{p0}^{\pm\epsilon} \quad (4.33)$$

solves the Dirac equation (4.19) with every summand $\sqrt{\rho_{pj}} e^{i\hbar\phi_{pj}}$. It computes the wave spinor $\psi_p^{\pm\epsilon}$ as the geometric projection of an initial relativistic spinor $\xi_{p0}^{\pm\epsilon}$ on a filter of direction $\mathbf{n}_p$ or as a solution of the relativistic eigenspinor equation (4.14) for $\psi_p^{\pm\epsilon} = \xi_p^{\pm\epsilon} = \xi_{p0}^{\pm\epsilon}$ in free space. (4.33) holds for a electron with positive constant energy E_{p+} and a positron with constant negative energy E_{p-} both with $s = 1$.

Consider now a positron with known relativistic momentum $\bar{\mathbf{p}}_-$ in (4.2) and negative energy E_- . Its eigenspinor $\xi^{\uparrow-}$ of (4.15) solves (4.33), and we consider the positron path travelling against time [22]. A plastic collision of the positron with a photon of momentum $\bar{\mathbf{p}}$ from §(c) leads to a new relativistic momentum $\bar{\mathbf{p}}_+ = \bar{\mathbf{p}} - \bar{\mathbf{p}}_-$. This in turn leads to an electron with known relativistic momentum $\bar{\mathbf{p}}_+$ in (4.2), whose energy E_+ is positive if the energy of the photon was larger than $E_0 - E_-$, with $E_0 = M_0 c^2$ the rest energy. The eigenspinor of the electron $\xi^{\uparrow+}$ of (4.15) now solves (4.33).

Although the above creation of electrons and positrons is a well-established Feynman diagram [22,67] in QED, the novelty is that it is now purely derived from the relativistic action of quaternions (4.32). Since theorem 3.2 applies to multiple particles, a second quantization is not needed to compute particle creation or annihilation phenomena. ■

5. Concluding remarks

This paper shows that the Schrödinger equation and its relativistic counterparts can be solved exactly from a $\mathbb{J}$ -valued action of theorem 2.4 and the classical densities along the associated multipaths. Identical wave functions and experimentally observed probability distributions can thus be obtained from three different interpretations:

- The Schrödinger (1.6), Klein-Gordon (4.5), Pauli (4.18), Dirac (4.19), and Maxwell (4.22) equations, which have no particle path until the wave function collapses at a measurement.
- The non-relativistic Feynman path integral (3.7), which has an ∞^∞ of time-sliced zig-zag paths with non-classical actions [2].
- Theorem 3.2, which uses the $\mathbb{J}$ -valued classical multipaths or actions from theorem 2.4 as a subset of all zig-zag paths in Feynman's path integral (3.7). The quantum probability matrix ϱ is generated from the initial classical density distribution ρ propagated along all classical determined paths of each action branch. Here the wave collapse at a measurement stems from the collapse of a classical density distribution to a Dirac impulse when it passes through a measurement device in lemma 3.3. At measurement time, the initial condition in $\mathbf{x}_0$ and $\mathbf{p}_0$ of theorem 2.4 is completed by the measurement $\mathbf{y}_k$, which determines in theorem 2.4 the $\mathbb{J}$ classical multipaths (2.5) from $\{\mathbf{x}_0 \vee \mathbf{p}_0\}$ to $\mathbf{y}_k$.

The fundamental quantum postulates on the existence of a wave function, its propagation with the Schrödinger equation in theorem 3.2 and the wave collapse at a measurement in lemma 3.3 are derived from the classical theorem 2.4. Furthermore, analytic computations of the classical action are simpler than solving the Feynman path integral and potentially easier than solving the Schrödinger equation directly. In addition, theorem 3.2 is a multi-particle result.

The $\mathbb{J}$ classical multipaths in theorem 3.2 and lemma 3.3 are strictly determined by the initial and final conditions. In the double slit experiment, the probabilistic quantum observation results from the non-Lipschitz constraint force in the slit. For the harmonic oscillator, the Coulomb wave,

the particle in the box, or the spinning particle, the initial probabilistic density distribution is classically propagated forward in time. In the EPR experiment [64,65], theorem 2.4 determines a constant angular momentum $\chi_o^\uparrow, \chi_o^\downarrow$ over time, and lemma 3.3 in turn allows a classical interpretation that the decision which spin correlation is sensed behind the filters is already taken when the particles separate.

Theorems 2.4 and 3.2 yield exact and purely classically-based derivations of well-known quantum wave functions, e.g. for the two-slit experiment (example 3.5), the Aharonov-Bohm effect (example 3.6), a particle in a box (example 3.7), quantum tunnelling (example 3.8), the hydrogen atom (example 3.10), a spinning particle and entanglement (example 4.1), and relativistic electron-positron creation (example 4.2).

Motivated by example 3.8, current research focuses on systematically deriving complex actions for general nonlinear potentials. This may enable exact wave computations where so far perturbation theory [44] had to be used. The action-based perspective may also simplify the study of systems modelled as composites of quantum and classical dynamics [68–70]. Exact quantum simulations may be obtained from classical physical systems with matching least action paths. The differentiability of the classical paths may make machine learning techniques readily applicable, e.g. in computational quantum chemistry. Finally, the ability to derive quantum quantities just from the $\mathbb{J}$ -valued classical action paths may have implications on some of the assumptions in quantum information processing.

Data accessibility. This article does not contain any additional data.

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References

1. Dirac P. 1933 The Lagrangian in quantum mechanics. *Phys. J. Soviet Union* **3**, 64–72.
2. Feynman R. 1948 Space-time approach to non-relativistic quantum mechanics. *Rev. Modern Phys.* **20**, 367–387. (doi:10.1103/RevModPhys.20.367)
3. Feynman R, Hibbs A. 1965 *Quantum mechanics and path integrals*. New York (Maidenhead, UK): McGraw-Hill.
4. Duru I, Kleinert H. 1982 Quantum Mechanics of H-atoms from path integrals. *Fortschritte Phys.* **30**, 401–435. (doi:10.1002/prop.19820300802)
5. Kleinert H. 2009 *Path integrals in quantum mechanics, statistics, polymer physics and financial markets*. Singapore: World Scientific.
6. Madelung E. 1927 Quantentheorie in hydrodynamischer Form. *Z. Phys.* **40**, 322–326. (doi:10.1007/BF01400372)
7. Bohm D. 1952 A suggested interpretation of the quantum theory in terms of 'hidden' variables. *Phys. Rev.* **85**, 166–179. (doi:10.1103/PhysRev.85.166)
8. Landau LD, Lifshitz EM. 1991 *Quantum mechanics: non-relativistic theory*, vol. 3 Course of Theoretical Physics. England: Pergamon Press.
9. Vleck JHV. 1928 The correspondence principle in the statistical interpretation of quantum mechanics. *Proc. Natl Acad. Sci. USA* **14**, 178–188. (doi:10.1073/pnas.14.2.178)
10. Maslov VP. 1972 *Théorie des perturbations et méthodes asymptotiques*. Moscow: Éditions Mir.
11. Berry MV, Mount KE. 1972 Semiclassical approximations in wave mechanics. *Rep. Prog. Phys.* **35**, 315–397. (doi:10.1088/0034-4885/35/1/306)
12. Groenewold HJ. 1946 On the principles of elementary quantum mechanics. *Physica* **12**, 405–460. (doi:10.1016/S0031-8914(46)80059-4)

13. Moyal JE. 1949 Quantum mechanics as a statistical theory. *Proc. Camb. Philos. Soc.* **45**, 99–124. (doi:10.1017/S0305004100000487)
14. Lagrange J. 1788 *Mécanique analytique*. Paris: Chez la veuve Desaint.
15. Lovelock D, Rund H. 1989 *Tensors, differential forms, and variational principles*. New York (Mineola, NY): Dover.
16. Goldstein H. 1980 *Classical mechanics*. Reading, MA (Wokingham, UK): Addison-Wesley.
17. Hamilton R. 1835 Second essay on a general method in dynamics. In *Philosophical Transactions of the Royal Society*.
18. Jacobi C. 1827 Über die Integration der partiellen Differentialgleichungen erster Ordnung. *J. Reine Angew. Math.* **1827**, 317. (doi:10.1515/crll.1827.2.317)
19. Beltrami E. 1902 *Ricerche di analisi applicata alla geometria*, vol. 1. Milano: Opere Matematiche.
20. Laplace PS. 1799 *Mécanique Céleste*. Paris: J.B.M. Duprat et al. Five volumes, 1799–1825.
21. Lorenz L. 1867 On the identity of the vibrations of light with electrical currents. *Philos. Mag.* **34**, 287–301. (doi:10.1080/14786446708639882)
22. Feynman RP. 1998 *Quantum electrodynamics*. MA: Addison-Wesley: Reading.
23. Euler L. 1755 Principes généraux du mouvement des fluides. *Acad. R. Sci. Belles Lett.* **11**, 274–315.
24. Planck M. 1901 Zur theorie des gesetzes der energieverteilung im normalspektrum. *Verh. Dtsch. Phys. Ges.* **2**, 237–245.
25. Cohen-Tannoudji C, Diu B, Laloe F. 2019 *Quantum mechanics*, 2nd edn. New York (Chichester, UK): Wiley.
26. Schrödinger E. 1926 Quantisierung als Eigenwertproblem. *Ann. Phys.* **384**, 361–376. (doi:10.1002/andp.19263840404)
27. Born M. 1926 Zur Quantenmechanik der Stoßvorgänge. *Z. Phys.* **37**, 863–867. (doi:10.1007/BF01397477)
28. Fulling S. 1996 *Aspects of quantum field theory in curved space-time*. Cambridge, UK: Cambridge University Press.
29. Gordon W. 1926 Der Compton Effekt nach der Schrödingerschen Theorie. *Z. Phys.* **40**, 117–133. (doi:10.1007/BF01390840)
30. Klein O. 1926 Quantentheorie und fünfdimensionale Relativitätstheorie. *Z. Phys.* **37**, 895–906. (doi:10.1007/BF01397481)
31. Pauli W. 1927 Zur Quantenmechanik des magnetischen Elektron. *Z. Phys.* **74**, 337–378.
32. Dirac PAM. 1928 The quantum theory of the electron. *Proc. R. Soc. Lond. A* **117**, 610–624. (doi:10.1098/rspa.1928.0023)
33. Maxwell JC. 1865 A dynamical theory of the electromagnetic field. *Phil. Trans. R. Soc. Lond.* **155**, 459–512. (doi:10.1098/rstl.1865.0008)
34. Picard E. 1890 Sur l'application des méthodes d'approximation successive à l'étude de certaines équations différentielles ordinaires. *J. Math. Pures Appl.* **6**, 145–210.
35. Lipschitz R. 1869 Über die Existenz einer Lösung von Differentialgleichungen. *Math. Ann.* **1**, 28–32.
36. Lohmiller W, Slotine JJ. 1998 On contraction analysis for nonlinear systems. *Automatica* **34**, 683–696. (doi:10.1016/S0005-1098(98)00019-3)
37. Lohmiller W, Slotine JJ. 2005 Contraction analysis of nonlinear distributed systems. *Int. J. Control* **78**, 678–688. (doi:10.1080/00207170500130952)
38. Newton I. 1687 *Philosophiæ naturalis principia mathematica*. London: Jussu Societatis Regiæ ac Typis Josephi Streater. Prostat apud plures bibliopolas.
39. Schleich WP, Greenberger DM, Kobe DH, Scully MO. 2013 Schrödinger equation revisited. *Proc. Natl Acad. Sci.* **110**, 5374–5379. (doi:10.1073/pnas.1302475110)
40. Pelster A, Wunderlin A. 1992 On the generalization of the Duru-Kleinert-propagator transformations. *Z. Phys. Condens. Matter* **89**, 373–386. (doi:10.1007/BF01318171)
41. Sakoda S. 2017 On the effective potential of Duru-Kleinert path integrals. *J. Math. Phys.* **58**, 062111. (doi:10.1063/1.4989986)
42. Fujikawa K. 1997 Path integral of the hydrogen atom, the Jacobi's principle of least action and one-dimensional quantum gravity. *Nuclear Phys.* **484**, 495–520. (doi:10.1016/S0550-3213(96)00584-6)
43. Bohr N. 1928 Das Quantenpostulat und die neuere Entwicklung der Atomistik. *Naturwissenschaften* **16**, 245–257. (doi:10.1007/BF01504968)
44. Heisenberg W. 1925 Über quantentheoretische Umdeutung kinematischer und mechanischer Beziehungen. *Z. Phys.* **33**, 879–893. (doi:10.1007/BF01328377)

45. Heisenberg W. 1958 *Physics and philosophy: the revolution in modern science*. London: George Allen and Unwin.
46. Bohr N. 1913 On the constitution of atoms and molecules. *Philos. Mag.* **26**, 1–25. (doi:10.1080/14786441308634955)
47. von Fraunhofer J. 1823 Neue Beobachtungen Über die Beugung des Lichts. *Annal. Phys.*
48. Zwiebach B. 2022 *Quantum mechanics*. Cambridge: The MIT Press.
49. Aharonov Y, Bohm D. 1959 Significance of electromagnetic potentials in the quantum theory. *Phys. Rev.* **115**, 485–491. (doi:10.1103/PhysRev.115.485)
50. Hund F. 1927 Zur Deutung der Molekülspektren. I. *Zeitschr. f. Physik. Z. Phys.* **40**, 742–764. (doi:10.1007/BF01400234)
51. Devoret MH, Martinis JM, Clarke J. 1985 Measurements of macroscopic quantum tunneling out of the zero-voltage state of a current-biased Josephson junction. *Phys. Rev. Lett.* **55**, 1908–1911. (doi:10.1103/PhysRevLett.55.1908)
52. Kepler J. 1609 *Astronomia Nova*. Prague: Typographia Sauriana.
53. Hilbert D. 1904 Grundzüge einer allgemeinen Theorie der linearen Integralgleichungen. *Nachrichten von der Gesellschaft der Wissenschaften zu Göttingen, Math.-Phys. Klasse* **1904**, 49–91.
54. Schrödinger E. 1926 An undulatory theory of the mechanics of atoms and molecules. *Phys. Rev.* **28**, 1049. (doi:10.1103/PhysRev.28.1049)
55. Einstein A. 1916 Die Grundlage der allgemeinen Relativitätstheorie. *Ann. Phys.* **354**, 769–822. (doi:10.1002/andp.19163540702)
56. Liboff R. 2002 *Introductory quantum mechanics*. Reading, MA (Wokingham, UK): Addison Wesley.
57. Einstein A. 1905 Zur Elektrodynamik bewegter Körper. *Ann. Phys.* **322**, 891–921. (doi:10.1002/andp.19053221004)
58. Lorentz HA. 1904 Electromagnetic phenomena in a system moving with any velocity smaller than that of light. *Proc. R. Netherlands Acad. Arts Sci.* **6**, 809–831.
59. Hawking SW. 1975 Particle creation by black holes. *Comm. Math. Phys.* **43**, 199–220. (doi:10.1007/BF02345020)
60. Bloch F. 1946 Nuclear induction. *Phys. Rev.* **70**, 460–474. (doi:10.1103/PhysRev.70.460)
61. Hamilton WR. 1828 Theory of systems of rays. *Trans. R. Irish Acad.* **15**, 69–174.
62. Einstein A, Podolsky B, Rosen N. 1935 Can the quantum-mechanical description of physical reality be considered complete? *Phys. Rev.* **47**, 777. (doi:10.1103/PhysRev.47.777)
63. Bell JS. 1964 On the Einstein Podolsky Rosen paradox. *Physics* **1**, 195–200. (doi:10.1103/PhysicsPhysiqueFizika.1.195)
64. Clauser JF, Horne MA, Shimony A, Holt RA. 1969 Proposed experiment to test local hidden-variable theories. *Phys. Rev. Lett.* **23**, 880–884. (doi:10.1103/PhysRevLett.23.880)
65. Aspect A. 1999 Bell's inequality test more ideal than ever. *Nature* **398**, 189–190. (doi:10.1038/18296)
66. Bouwmeester D, Ekert A, Zeilinger A. 2000 *The physics of quantum information: quantum cryptography, quantum teleportation, quantum computation*. Berlin, Heidelberg: Springer.
67. Weinberg S. 1964 Feynman rules for any spin. *Phys. Rev.* **133**, B1318–B1332. (doi:10.1103/PhysRev.133.B1318)
68. Hooft G't. 2007 A mathematical theory for deterministic quantum mechanics. *J. Phys: Conf. Ser.* **67**, 012015. (doi:10.1088/1742-6596/67/1/012015)
69. Dalibard J, Castin Y, Dalibard J. 1993 Monte-Carlo wave function method in quantum optics. *J. Opt. Soc. Am. B* **10**, 524–538. (doi:10.1364/JOSAB.10.000524)
70. Wang F, Makri N. 2019 Quantum-classical path integral. *J. Chem. Phys.* **150**, 184102. (doi:10.1063/1.5091725)